Measuring Normative Information

What a value-alignment dataset can and cannot tell you, derived from zero

Ivan GAN

ivan@ivan.icu www.ivan.icu github.com/ivanicu

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Abstract

A public dataset of human value judgments ships 15,248 written criteria with numerical weights, and 82.3% of those criteria contain no condition, no exception, no priority, no hedge and no prohibition – the collected normative payload is content, a sign, and a magnitude. This paper asks what can be measured about such an object at all, and answers it as a typed question: every quantity is defined relative to a declared query family, a declared decision family, and a named instrument, and each is derived here from nothing, assuming no statistics. What survives is: (i) a theorem that no finite weight encodes a veto, because the penalty reproducing one depends on which candidate responses happened to be shown, with the counterexample present in the data at 2.65× the elicitation scale; (ii) that a person's own written criteria predict a *stranger's* ranking at 0.310 and their own at 0.393, so the individual signal in the elicitation is 0.083 of cosine and nothing downstream can destroy what was never collected; (iii) that comparative measurements are invariant across three executors and a presentation-order swap (Spearman 0.963–1.000) while absolute ones are not (40% of winners change with the reader, 8 points with the page order); and (iv) that one bit per criterion – the sign alone – reproduces 89.8% of the shipped decisions. Six claims were withdrawn during this work and are listed with what killed them; five of the six died the same way, to a verdict decided by a constant the author chose rather than by a comparison against a range of defensible references. The central claim ends if a person's induced ordering proves unstable across response sets: what was elicited would then be a property of the probe rather than of the person, and no measurement downstream would be about values at all.

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1 Introduction

1.1 The question, and why the obvious version of it is the wrong one

The obvious question is: *how much of what people said survives the pipeline?* It has no answer, and the reason is not empirical.

To say that two normative objects carry "the same information" one must fix what one intends to ask of them. Proposition 97 shows that the choice of question family determines the answer completely: under one admissible family every change is total loss, under another no change is any loss. A framework that reports a single percentage has therefore not measured a property of the pipeline; it has reported a choice it did not disclose.

The question this paper answers instead is: *given a declared family of decisions, what is identified, what is bounded, and what is forbidden?* That version has answers, several of them are theorems rather than measurements, and the largest of them turns out to concern the elicitation rather than the pipeline.

1.2 Scope, identity, and contributions

The object is a released dataset of human comparative judgments over four candidate responses per prompt, together with written criteria and numeric weights, and a compiled "core" standard derived from them (§4). The satisfaction of each criterion by each response is not in the release; it is reconstructed with a language-model judge, and every consequence of that is tracked explicitly (§9).

Contributions, in the order they appear:

1. A self-contained derivation of every statistical tool used, from the definition of probability through the design effect, attenuation and minimum detectable effect (§2). No external statistical background is assumed at any point.
2. The probe bound (theorem 64): the elicitation returns two free parameters per person per prompt, whatever the criteria say.
3. A measurement that the weight scale is used as an *interval* scale, with the fraction of decisions that depend on it (§5.2).
4. The social-choice structure of the corpus, including the finding that the weighted-criteria route agrees with the Condorcet winner only 68.4% of the time while every ranking rule agrees 97–98% (§6).
5. theorem 80 and theorem 81: the exact condition under which a penalty reproduces a veto, and the proof that no finite penalty does so uniformly.
6. Normalised regret (definition 99) as the decision-theoretically correct estimand, with the exact identity showing the flip rate to be its coarsening (lemma 101).
7. The gauge result separating comparative from absolute claims (§9.3).
8. An explicit non-identification section with proofs (§15), and a design specification derived line by line from it (§21).

1.3 Relation to neighbouring work

Everything in §2 is standard and is reproduced only so the paper is self-contained. Blackwell's comparison of experiments, Arrow's impossibility theorem, Sen's liberal paradox, the Shapley value, the Spearman–Brown prophecy formula, the design effect and classical attenuation are all classical; they are derived rather than cited because a reader with no statistical background cannot otherwise check the argument. The novelty boundary is stated explicitly in §17 and the Discussion: what is new is the application to this object and five specific measured quantities, not the machinery.

1.4 Roadmap

§2 builds the mathematics. §4 defines the object exactly. §12.1 states the measurement contract. §5–§9 walk the chain one arrow at a time. §13 and §14 give the estimators. §15 proves what cannot be identified. §16 states the findings with their confidence cards, §17 what did not survive, §18 what would end each claim, and §21 what a sufficient design requires.

2 The mathematics, from zero

This section assumes no statistics. It builds, in order, every object the rest of the paper uses. Nothing here is specific to the dataset; §4 onward uses only what is defined below. A reader who already knows this material should still read §2.13, §2.12 and §2.17, because those three are where papers in this area go wrong.

2.1 Sets, functions, and the two symbols that do all the work

A *set* is a collection of distinct things, written $\{a, b, c\}$. We write $x \in \mathcal{A}$ for "x is a member of $\mathcal{A}$", and $|\mathcal{A}|$ for the number of members.

A *function* $f : \mathcal{A} \to \mathcal{B}$ assigns to every member of $\mathcal{A}$ exactly one member of $\mathcal{B}$.

Two symbols carry most of the load.

- $\sum_{i=1}^{n} x_i$ means $x_1 + x_2 + \cdots + x_n$. It is an instruction to add. Nothing more.
- $\mathbf{1}\{P\}$ is the *indicator*: it equals 1 if the statement P is true and 0 if false. It converts a yes/no fact into a number, and it is how every "what fraction of cases" quantity in this paper is built.

The single most useful identity in the paper follows immediately from those two: the average of an indicator *is* a proportion.

$$\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{P_i\} \;=\; \frac{\text{number of } i \text{ with } P_i \text{ true}}{n}. \tag{1}$$

Every rate reported in this paper – how often a decision changes, how often a veto binds – is an instance of equation (1). That is why they can all be treated with one set of tools.

2.2 Finite probability

Definition 1 (Probability space)**.** A *finite probability space* is a finite set Ω of *outcomes* together with a function $\mathbb{P}$ assigning to each $\omega \in \Omega$ a number $\mathbb{P}(\omega) \geq 0$, such that $\sum_{\omega\in\Omega}\mathbb{P}(\omega) = 1$. For a subset $E \subseteq \Omega$ (an *event*), $\mathbb{P}(E) := \sum_{\omega\in E}\mathbb{P}(\omega)$.

Two consequences, both immediate from the definition and both used later.

Sub-step 0.2.1 — probabilities are bounded.

$E \subseteq \Omega$ implies $\mathbb{P}(E) \leq \sum_{\omega\in\Omega}\mathbb{P}(\omega) = 1$, and $\mathbb{P}(E) \geq 0$ because it is a sum of non-negative terms. So $0 \leq \mathbb{P}(E) \leq 1$ always.

Sub-step 0.2.2 — disjoint events add.

If $E \cap F = \varnothing$ then no outcome is counted twice, so $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F)$.

Definition 2 (Random variable)**.** A *random variable* is a function $X : \Omega \to \mathbb{R}$. It is not random and it is not a variable; it is a rule that reads off a number from whatever outcome occurred. We write $\mathbb{P}(X = x)$ for $\mathbb{P}(\{\omega : X(\omega) = x\})$.

2.3 Expectation: the derivation, not the formula

Suppose we observe X many times and average the results. If outcome x occurs a fraction $\mathbb{P}(X = x)$ of the time, then in n observations it contributes $x \cdot n\mathbb{P}(X = x)$ to the total. Summing over the possible values and dividing by n:

$$\frac{1}{n}\sum_{\text{values } x} x \cdot n\,\mathbb{P}(X = x) \;=\; \sum_{x} x\,\mathbb{P}(X = x). \tag{2}$$

The n cancels. The result does not depend on how many times we look. That limit-of-the-average is the definition.

Definition 3 (Expectation)**.** $\mathbb{E}[X] := \sum_{x} x\,\mathbb{P}(X = x)$.

Lemma 4 (Linearity)**.** *For random variables X, Y on the same space and constants a, b:* $\mathbb{E}[aX+bY] = a\mathbb{E}[X]+b\mathbb{E}[Y]$.

Proof. Write the expectation as a sum over outcomes rather than values, which is equivalent because grouping outcomes by their value only reorders a finite sum: $\mathbb{E}[X] = \sum_{\omega} X(\omega)\mathbb{P}(\omega)$. Then

$$\mathbb{E}[aX + bY] = \sum_{\omega}\big(aX(\omega) + bY(\omega)\big)\mathbb{P}(\omega) = a\sum_{\omega}X(\omega)\mathbb{P}(\omega) + b\sum_{\omega}Y(\omega)\mathbb{P}(\omega),$$

which is $a\mathbb{E}[X]+b\mathbb{E}[Y]$. *No independence was used.* This is why linearity is the one tool that never needs an assumption, and it is used at every later step where an assumption would be contentious. □

Corollary 5 (Expectation of an indicator)**.** $\mathbb{E}[\mathbf{1}\{P\}] = 1 \cdot \mathbb{P}(P) + 0 \cdot (1 - \mathbb{P}(P)) = \mathbb{P}(P)$.

Corollary 5 is the bridge between "a proportion I measured" and "a probability I want to talk about", and every rate in this paper crosses it.

2.4 Variance: why squared deviation, and not something else

We want a single number for "how spread out is X". Let $\mu := \mathbb{E}[X]$.

Sub-step 0.4.1 — the naive attempt fails.

$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0$ by lemma 4, always. Average deviation is identically zero and carries no information: positive and negative deviations cancel exactly.

Sub-step 0.4.2 — two repairs are available.

Remove the sign either by absolute value, $\mathbb{E}|X - \mu|$, or by squaring, $\mathbb{E}[(X - \mu)^2]$. Both are legitimate measures of spread. Squaring is chosen for one reason that matters here and one that does not.

- **Matters:** squared deviations *add* for independent quantities (lemma 12), so the spread of an average has a closed form. Absolute deviations do not add.
- **Does not matter:** differentiability, which is a convenience for optimisation and irrelevant to this paper.

Definition 6 (Variance and standard deviation)**.** $\mathrm{Var}(X) := \mathbb{E}[(X-\mu)^2]$ and $\mathrm{sd}(X) := \sqrt{\mathrm{Var}(X)}$. The square root exists because $\mathrm{Var}(X)$ is an average of squares, hence ≥ 0. The standard deviation is in the same units as X; the variance is in squared units, which is why intervals are reported with the former.

Lemma 7 (Computational form)**.** $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Proof. $\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2$, using lemma 4 and the fact that μ is a constant. □

Corollary 8 (Variance of an indicator)**.** *If $X = \mathbf{1}\{P\}$ with $\mathbb{P}(P) = \pi$, then $X^2 = X$ (since $0^2 = 0, 1^2 = 1$), so by lemma 7* $\mathrm{Var}(X) = \pi - \pi^2 = \pi(1-\pi)$.

Corollary 8 is why every yes/no rate in this paper has a variance we can write down without estimating anything extra, and why rates near 0 or 1 are measured more precisely than rates near $\frac{1}{2}$.

Lemma 9 (Scaling)**.** $\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X)$.

Proof. $\mathbb{E}[aX + b] = a\mu + b$, so the deviation is $aX + b - (a\mu + b) = a(X - \mu)$, whose square is $a^2(X-\mu)^2$. Take expectations and apply lemma 4. The additive constant b vanishes because shifting everything does not change spread. □

2.5 Independence, covariance, correlation

Definition 10 (Independence)**.** X and Y are *independent* if $\mathbb{P}(X = x \text{ and } Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$ for all x, y: knowing one tells you nothing about the other.

Definition 11 (Covariance)**.** $\mathrm{Cov}(X,Y) := \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$.

The sign of the product $(X-\mu_X)(Y-\mu_Y)$ is positive when both deviate the same way and negative when they deviate oppositely, so the average is positive for co-movement and negative for counter-movement. Note $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$.

Lemma 12 (Variance of a sum)**.** $\mathrm{Var}(X+Y) = \mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$. *If X, Y are independent then* $\mathrm{Cov}(X,Y) = 0$ *and the variances simply add.*

Proof. Let $D_X := X - \mu_X$, $D_Y := Y - \mu_Y$. Then $X + Y - (\mu_X + \mu_Y) = D_X + D_Y$, and $(D_X + D_Y)^2 = D_X^2 + 2D_X D_Y + D_Y^2$. Take expectations, apply lemma 4, and read off the three terms. For the second claim: under independence $\mathbb{E}[D_X D_Y] = \mathbb{E}[D_X]\mathbb{E}[D_Y] = 0 \cdot 0 = 0$. □

Remark 13 (The converse is false, and it matters here). $\mathrm{Cov}(X,Y) = 0$ does *not* imply independence. Covariance detects only linear co-movement. A quantity can be completely determined by another and still have zero covariance with it. Every "uncorrelated, therefore unrelated" inference in this literature is invalid for this reason, and §14 uses a *non-linear* measure of dependence precisely to avoid it.

Theorem 14 (Cauchy–Schwarz)**.** $|\,\mathrm{Cov}(X,Y)| \leq \mathrm{sd}(X)\,\mathrm{sd}(Y)$.

Proof. Write D_X, D_Y as above and let $t \in \mathbb{R}$. Since a square is never negative,

$$0 \;\leq\; \mathbb{E}\big[(D_X + tD_Y)^2\big] \;=\; \mathbb{E}[D_X^2] + 2t\,\mathbb{E}[D_X D_Y] + t^2\,\mathbb{E}[D_Y^2] \;=\; \mathrm{Var}(X) + 2t\,\mathrm{Cov}(X,Y) + t^2\,\mathrm{Var}(Y).$$

The right-hand side is a quadratic in t that is never negative. A quadratic $at^2 + bt + c$ with $a \geq 0$ is non-negative for all t if and only if its discriminant $b^2 - 4ac \leq 0$ (otherwise it has two distinct real roots and dips below zero between them). Here $a = \mathrm{Var}(Y)$, $b = 2\,\mathrm{Cov}(X,Y)$, $c = \mathrm{Var}(X)$, so

$$4\,\mathrm{Cov}(X,Y)^2 - 4\,\mathrm{Var}(X)\,\mathrm{Var}(Y) \leq 0 \implies \mathrm{Cov}(X,Y)^2 \leq \mathrm{Var}(X)\,\mathrm{Var}(Y).$$

Take square roots. □

Definition 15 (Correlation)**.** If both standard deviations are non-zero,

$$\operatorname{Corr}(X,Y) \;=\; r \;:=\; \frac{\operatorname{Cov}(X,Y)}{\operatorname{sd}(X)\,\operatorname{sd}(Y)}.$$

Corollary 16. $-1 \le r \le 1$, *immediately from theorem 14.*

So r is covariance with the units divided out. It is *not* a percentage, it is *not* a share of anything, and $r = 0.5$ is not "half". The only quantity with a share interpretation is r^2, and only under a linear model.

2.6 Samples, estimators, and the thing that is actually random

We never see $\mathbb{P}$. We see n observations $x_1, \dots, x_n$ and must guess properties of $\mathbb{P}$ from them.

Definition 17 (Estimator)**.** An *estimator* is any function of the observed data used to guess a fixed unknown quantity (the *estimand*). We write $\hat{\theta}$ for an estimator of θ.

Definition 18 (Bias)**.** $\operatorname{Bias}(\hat{\theta}) := \mathbb{E}[\hat{\theta}] - \theta$. An estimator is *unbiased* if this is zero: it is right *on average across repetitions of the whole study*, which says nothing about any single study.

Remark 19 (What is random). θ is a fixed number. $\hat{\theta}$ varies because the sample varies. Every statement below about "the distribution of $\hat{\theta}$" is a statement about what would happen across hypothetical repetitions of the study, not about θ. This is the single most misread point in applied statistics and it is the reason §2.10 states explicitly what a confidence interval does *not* mean.

2.7 The sample mean, and the derivation of the standard error

Let $X_1, \dots, X_n$ be independent, each with mean μ and variance σ^2. Define $\bar{X} := \frac{1}{n}\sum_i X_i$.

Sub-step 0.7.1 — the mean is unbiased.

By lemma 4, $\mathbb{E}[\bar{X}] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu$.

Sub-step 0.7.2 — the variance of the mean.

By lemma 9 with $a = 1/n$, then lemma 12 with independence:

$$\operatorname{Var}(\bar{X}) \;=\; \frac{1}{n^2}\operatorname{Var}\Big(\sum_i X_i\Big) \;=\; \frac{1}{n^2}\sum_i \operatorname{Var}(X_i) \;=\; \frac{n\sigma^2}{n^2} \;=\; \frac{\sigma^2}{n}. \tag{3}$$

Sub-step 0.7.3 — naming it.

Definition 20 (Standard error)**.** $\operatorname{se}(\bar{X}) := \operatorname{sd}(\bar{X}) = \sigma/\sqrt{n}$.

Remark 21 (The $\sqrt{n}$ is the whole economics of measurement). Halving the standard error requires *four times* the data. This single fact determines every sample-size statement in §2.12, and it is why the question "how many prompts do we need" has an answer that is often unaffordable.

Remark 22 (Independence is doing real work in equation (3)). If the X_i are positively correlated, lemma 12 adds covariance terms and the true variance is *larger* than σ^2/n. Using equation (3) anyway understates the uncertainty. §2.13 quantifies exactly how much, because this specific error occurred in this work.

2.8 Why $n - 1$

To use $\operatorname{se} = \sigma/\sqrt{n}$ we need σ^2, which we also do not know. The obvious estimator $\frac{1}{n}\sum_i (X_i - \bar{X})^2$ is biased downward, and the reason is worth the derivation because it exposes what "degrees of freedom" means.

Lemma 23. $\mathbb{E}\Big[\sum_{i=1}^{n}(X_i - \bar{X})^2\Big] = (n-1)\sigma^2.$

Proof. Expand around the true mean μ by adding and subtracting it:

$$\sum_i (X_i - \bar{X})^2 = \sum_i \big((X_i - \mu) - (\bar{X} - \mu)\big)^2 = \sum_i (X_i - \mu)^2 - 2(\bar{X} - \mu)\sum_i (X_i - \mu) + n(\bar{X} - \mu)^2.$$

Now $\sum_i (X_i - \mu) = n(\bar{X} - \mu)$, so the middle term is $-2n(\bar{X} - \mu)^2$ and the last two combine to $-n(\bar{X} - \mu)^2$:

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Take expectations. The first term is $n\sigma^2$. For the second, $\mathbb{E}[(\bar{X} - \mu)^2] = \mathrm{Var}(\bar{X}) = \sigma^2/n$ by equation (3), so it contributes $n \cdot \sigma^2/n = \sigma^2$. Hence the expectation is $n\sigma^2 - \sigma^2 = (n-1)\sigma^2$. □

Definition 24 (Sample variance)**.** $s^2 := \dfrac{1}{n-1} \sum_i (X_i - \bar{X})^2$, which by lemma 23 satisfies $\mathbb{E}[s^2] = \sigma^2$.

The interpretation.

Deviations are taken from $\bar{X}$, which was computed from the same data and therefore sits closer to the points than μ does. One degree of freedom was spent estimating the centre. Dividing by $n-1$ returns it.

2.9 The central limit theorem, and precisely what it licenses

Theorem 25 (Central limit theorem, stated)**.** *If $X_1, \ldots, X_n$ are independent and identically distributed with mean μ and finite variance σ^2, then as n grows the distribution of $(\bar{X} - \mu)/(\sigma/\sqrt{n})$ approaches the standard normal distribution: $\mathbb{P}(Z \le z) \to \Phi(z)$ where $\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\, dt$.*

We do not prove this. What matters is what it does and does not say.

- **Says:** the *average* becomes normally distributed regardless of the shape of the individual observations. This is why a rate built from 0/1 indicators – about as non-normal as data gets – still has a normal sampling distribution.
- **Does not say:** that the data are normal. They are not, and it does not matter.
- **Fails when:** observations are dependent (§2.13), or n is small with a skewed distribution, or the quantity is not an average. In this paper the third case arises for correlations, which is why §2.14 uses resampling instead of theorem 25 there.

Three quantiles of Φ are used throughout and are quoted once here:

$$z_{0.975} = 1.9600, \qquad z_{0.80} = 0.8416, \qquad z_{0.99} = 2.3263. \tag{4}$$

z_q is defined by $\Phi(z_q) = q$: the point with q of the distribution below it.

2.10 Confidence intervals: derivation, and what they do not mean

Sub-step 0.10.1.

By theorem 25, with probability 0.95,

$$-1.96 \;\le\; \frac{\bar{X} - \mu}{\mathrm{se}} \;\le\; 1.96.$$

Sub-step 0.10.2.

Multiply through by se (positive, so inequalities are preserved): $-1.96\ \mathrm{se} \le \bar{X} - \mu \le 1.96\ \mathrm{se}$.

Sub-step 0.10.3.

Subtract $\bar{X}$ and multiply by -1, which *reverses* both inequalities: $\bar{X} - 1.96\ \mathrm{se} \le \mu \le \bar{X} + 1.96\ \mathrm{se}$.

Definition 26 (95% confidence interval)**.** $\left[\bar{X} - 1.96\ \mathrm{se},\ \bar{X} + 1.96\ \mathrm{se}\right]$.

Remark 27 (What it means). Across hypothetical repetitions of the entire study, 95% of the intervals so constructed would contain the fixed unknown μ. The randomness is in the interval, not in μ.

Remark 28 (What it does not mean). It does *not* mean $\mathbb{P}(\mu \in [\ell, u]) = 0.95$ for the particular interval computed: μ is a constant and either lies in it or does not. It does *not* mean 95% of future observations fall in it – that is a prediction interval and is much wider. And an interval that excludes zero does not mean the effect is large; §2.12 separates those two questions.

2.11 Hypothesis tests, z-statistics, and p-values

Definition 29 (Null hypothesis)**.** A fully specified statement about the estimand, denoted H_0, typically $\theta = 0$. It is specified in advance so that the distribution of the estimator *under* it is known.

Definition 30 (z-statistic)**.** $z := \hat{\theta}/\mathrm{se}(\hat{\theta})$: the estimate measured in units of its own uncertainty. A z of 3 means the observed effect is three standard errors from zero.

Definition 31 (p-value)**.** $p := \mathbb{P}(|Z| \geq |z| \mid H_0)$, the probability of seeing an effect at least this extreme *if the null were true*.

Remark 32 (The three standard misreadings, each false). p is not $\mathbb{P}(H_0 \mid \text{data})$ – it is conditioned the other way round. p is not the probability the result is a fluke. And $1 - p$ is not the probability the effect is real. The p-value is a statement about data under an assumption, never about the assumption given data.

Remark 33 (Why this paper mostly reports z and intervals). A p-value collapses effect size and precision into one number, so a tiny effect measured precisely and a large effect measured sloppily can produce the same p. The interval keeps them separate, which is why every finding in §16 carries one.

2.12 Power, and the minimum detectable effect

A null result is uninterpretable without knowing what the design could have seen.

Definition 34 (Power)**.** The probability of rejecting H_0 when a specific alternative $\theta = \delta$ is true. Conventionally $1 - \beta = 0.80$.

Theorem 35 (Minimum detectable effect)**.** *For a two-sided test at level α with power $1 - \beta$, the smallest true effect detectable with that power is*

$$\mathrm{MDE} \;=\; \left(z_{1-\alpha/2} + z_{1-\beta}\right) \cdot \mathrm{se}\,.$$

Proof. Sub-step 1. We reject when $|\hat{\theta}| \geq z_{1-\alpha/2}\,\mathrm{se}$, by construction of the level-α test.

Sub-step 2. Suppose the truth is $\theta = \delta > 0$. Then $\hat{\theta}$ is centred at δ with standard deviation se, so $(\hat{\theta} - \delta)/\mathrm{se}$ is standard normal.

Sub-step 3. Rejection (ignoring the negligible wrong-tail probability) requires $\hat{\theta} \geq z_{1-\alpha/2}\,\mathrm{se}$, i.e.

$$\frac{\hat{\theta} - \delta}{\mathrm{se}} \;\geq\; z_{1-\alpha/2} - \frac{\delta}{\mathrm{se}}.$$

Sub-step 4. That event has probability $1 - \beta$ exactly when the threshold on the right equals $-z_{1-\beta}$, because $\mathbb{P}(Z \geq -z_{1-\beta}) = 1 - \beta$ by symmetry of the normal.

Sub-step 5. So $z_{1-\alpha/2} - \delta/\mathrm{se} = -z_{1-\beta}$, giving $\delta = (z_{1-\alpha/2} + z_{1-\beta})\,\mathrm{se}$. □

Corollary 36 (The number used throughout)**.** *At $\alpha = 0.05$, $1 - \beta = 0.80$: from equation (4),* $\mathrm{MDE} = (1.9600 + 0.8416)\ \mathrm{se} = 2.8016\ \mathrm{se}$.

Corollary 37 (Required sample size)**.** *Since* $\mathrm{se} = \sigma/\sqrt{n}$*, detecting an effect of size δ needs* $n = \left(2.8016\,\sigma/\delta\right)^2$.

Remark 38 (Reading a null). If MDE > |the effect the null was supposed to rule out|, the null is *silence*, not evidence of absence. §16 reports an MDE beside every null for this reason.

2.13 Clustering, the intraclass correlation, and the design effect

This subsection exists because the error it describes was made in this work and had to be corrected.

The setting.

Suppose the data have two levels: P independent *clusters* (here: prompts), and within each cluster R observations (here: repeated perturbations of the same prompt). Observations in the same cluster are *not* independent.

Sub-step 0.13.1 — the model.

Write $X_{pr} = \mu + a_p + e_{pr}$, where a_p is a cluster effect with variance σ_b^2 ("between"), e_{pr} is an observation effect with variance σ_w^2 ("within"), and all terms are independent of each other.

Sub-step 0.13.2 — the cluster mean.

$\bar{X}_p = \mu + a_p + \bar{e}_p$, and by lemma 12 and equation (3), $\mathrm{Var}(\bar{X}_p) = \sigma_b^2 + \sigma_w^2/R$.

Sub-step 0.13.3 — the overall mean.

The P cluster means *are* independent, so equation (3) applies to them:

$$\operatorname{Var}(\hat{\pi}) \;=\; \frac{1}{P}\Big(\sigma_b^2 + \frac{\sigma_w^2}{R}\Big). \tag{5}$$

Sub-step 0.13.4 — what the naive formula would give.

Treating all PR observations as independent with total variance $\sigma^2 = \sigma_b^2 + \sigma_w^2$ gives $\sigma^2/(PR)$.

Definition 39 (Intraclass correlation)**.** $\rho := \dfrac{\sigma_b^2}{\sigma_b^2 + \sigma_w^2}$, the share of total variance that is between clusters. Equivalently, the correlation between two observations drawn from the same cluster.

Theorem 40 (Design effect)**.**

$$\text{DEFF} \;:=\; \frac{\textit{correct variance equation (5)}}{\textit{naive variance}} \;=\; 1 + (R-1)\rho.$$

Proof. Divide equation (5) by $\sigma^2/(PR)$:

$$\frac{\frac{1}{P}(\sigma_b^2 + \sigma_w^2/R)}{\frac{\sigma_b^2+\sigma_w^2}{PR}} = \frac{R\sigma_b^2 + \sigma_w^2}{\sigma_b^2 + \sigma_w^2} = \frac{R\sigma_b^2 + \sigma_w^2 + (\sigma_b^2 - \sigma_b^2)}{\sigma_b^2 + \sigma_w^2} = 1 + \frac{(R-1)\sigma_b^2}{\sigma_b^2 + \sigma_w^2} = 1 + (R-1)\rho. \qquad \square$$

Corollary 41. *Standard errors computed as if the PR observations were independent are too small by a factor $\sqrt{1 + (R-1)\rho}$, and z-statistics are too large by the same factor.*

Remark 42 (Effective sample size). $n_{\text{eff}} = PR/\text{DEFF}$. With $R = 20$ and $\rho = 0.27$, DEFF $= 6.13$, so 19,360 rows carry the information of about 3,150 – and the honest count of independent units remains $P = 968$.

2.14 The bootstrap

The formulas above give se for a *mean*. For a correlation, a median, a ratio, or anything else, no such formula is available in general. The bootstrap replaces it with computation.

Protocol 43 (Cluster bootstrap)**.** Given clusters $c_1, \ldots, c_P$ and a statistic T:
1. Draw P clusters *with replacement* from $\{c_1, \ldots, c_P\}$.
2. Compute T on the resampled collection; call it $T^{(1)}$.
3. Repeat B times, obtaining $T^{(1)}, \ldots, T^{(B)}$.
4. Report the 2.5th and 97.5th percentiles as a 95% interval.

Why it works, in one line.

The variation of T across resamples of the observed data mimics the variation of T across fresh samples from the population, because the observed sample is our best available stand-in for that population.

Why the resampling unit must be the cluster.

Resampling *rows* would create datasets whose within-cluster dependence differs from the real one, and would reproduce the understatement of theorem 40. Every interval in this paper resamples prompts.

When it fails.

With few clusters, at the boundary of a parameter's range, or for statistics that are not smooth functions of the data. All three are checked in §20.

2.15 Permutation tests, and exactly what they test

Protocol 44 (Permutation test)**.** To test whether the *pairing* between two vectors x and y carries information: compute $T(x, y)$; then repeatedly shuffle x and recompute $T(x_{\text{shuffled}}, y)$; the p-value is the fraction of shuffles reaching $|T|$ at least as large as observed.

Remark 45 (The exchangeability assumption). Shuffling assumes that under H_0 all orderings are equally likely. If x has internal structure – clustering, time order – shuffling destroys that too, and the test then answers a question nobody asked. In §16 the shuffle is applied to cluster-level values for this reason.

Remark 46 (Resolution floor). With B shuffles the smallest attainable p is $1/(B+1)$: a reported $p = 0.0000$ from $B = 200$ means only $p < 0.005$, and no more.

Remark 47 (What a permutation null cannot do). It answers *did the pairing matter*, never *why*. It is a falsifier, never a confirmer of a mechanism.

2.16 Multiplicity

Theorem 48 (Bonferroni)**.** *If C tests are each performed at level α/C, then the probability of at least one false rejection when all nulls are true is at most α.*

Proof. Let A_i be the event of a false rejection on test i, so $\mathbb{P}(A_i) = \alpha/C$. The union bound gives $\mathbb{P}(\bigcup_i A_i) \le \sum_i \mathbb{P}(A_i) = C \cdot \alpha/C = \alpha$. The union bound itself follows from corollary 5 plus $\mathbf{1}\{\bigcup A_i\} \le \sum_i \mathbf{1}\{A_i\}$, which holds pointwise because the left side is 0 or 1 and the right side counts. □

Corollary 49 (The bar used in this paper)**.** *For $C \approx 170$ tests at $\alpha = 0.05$: $\alpha/C \approx 0.0003$, whose two-sided normal quantile is $|z| \approx 3.6$. This paper uses the more conservative $|z| > 3.9$ and reports the number of tests and the number of survivors together, since reporting only survivors is the multiplicity failure with manners.*

Remark 50 (Bonferroni versus false discovery rate). Bonferroni controls the probability of *any* false positive, which is the right target when a single false claim is costly. The Benjamini–Hochberg procedure instead controls the expected *share* of false positives among rejections, and rejects hypothesis (k) when $p_{(k)} \le qk/C$ for the largest such k. Its threshold at rank $k = C$ is q itself, not q/C – confusing the two demands orders of magnitude more resolution than BH actually requires.

2.17 Measurement error and attenuation

Every quantity in §4 is measured by a fallible instrument. This subsection derives the exact consequence.

Sub-step 0.17.1 — the model.

Suppose the truth is T but we observe $X = T + \varepsilon$, with ε independent of T, $\mathbb{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma_\varepsilon^2$.

Sub-step 0.17.2 — reliability.

Define

$$\mathrm{rel} \ := \ \frac{\mathrm{Var}(T)}{\mathrm{Var}(X)} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\varepsilon^2} \ \in \ [0,1],$$

the share of observed variance that is signal. Lemma 12 with independence gives the denominator.

Sub-step 0.17.3 — two independent measurements.

If $X_1 = T + \varepsilon_1$ and $X_2 = T + \varepsilon_2$ with $\varepsilon_1 \perp \varepsilon_2$ and equal error variances, then $\mathrm{Cov}(X_1, X_2) = \mathrm{Var}(T) = \sigma_T^2$ (the error terms contribute nothing), and each has variance $\sigma_T^2 + \sigma_\varepsilon^2$. Therefore

$$\mathrm{Corr}(X_1, X_2) \ = \ \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\varepsilon^2} \ = \ \mathrm{rel}. \tag{6}$$

The correlation between two independent measurements of the same thing <u>is</u> the reliability of one of them. This is what makes §9 computable at zero extra cost.

Theorem 51 (Attenuation)**.** *Let $X = T + \varepsilon_X$ and $Y = U + \varepsilon_Y$ with independent errors. Then*

$$\mathrm{Corr}(X,Y) \ = \ \mathrm{Corr}(T,U) \cdot \sqrt{\mathrm{rel}_X\,\mathrm{rel}_Y}.$$

Proof. $\mathrm{Cov}(X,Y) = \mathrm{Cov}(T + \varepsilon_X, U + \varepsilon_Y) = \mathrm{Cov}(T,U)$, since all three cross terms involve an error independent of everything else. The denominators are $\mathrm{sd}(X) = \sigma_T/\sqrt{\mathrm{rel}_X}$ and likewise for Y, because $\mathrm{rel}_X = \sigma_T^2/\sigma_X^2$ rearranges to $\sigma_X = \sigma_T/\sqrt{\mathrm{rel}_X}$. Substituting,

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(T,U)}{\frac{\sigma_T}{\sqrt{\mathrm{rel}_X}} \cdot \frac{\sigma_U}{\sqrt{\mathrm{rel}_Y}}} = \frac{\mathrm{Cov}(T,U)}{\sigma_T \sigma_U} \sqrt{\mathrm{rel}_X \mathrm{rel}_Y}.$$ □

Corollary 52 (The ceiling)**.** *A perfect true relationship,* $\mathrm{Corr}(T,U) = 1$*, reads at most $\sqrt{\mathrm{rel}_X \mathrm{rel}_Y}$. No amount of data raises this: it is bias, not noise, and $\sqrt{n}$ does not touch it.*

2.18 Reliability of an aggregate, and the Spearman–Brown formula

Averaging several noisy measurements reduces error. The exact statement:

Theorem 53 (Spearman–Brown)**.** *If a score is the sum of m exchangeable components each with reliability correlation r to another such component, then doubling the number of components gives reliability* $2r/(1+r)$.

Derivation. Let each component be $T + \varepsilon_i$ with $\mathrm{Var}(T) = \sigma_T^2$, $\mathrm{Var}(\varepsilon_i) = \sigma_\varepsilon^2$, errors independent. The sum of m components is $mT + \sum_i \varepsilon_i$, with signal variance $m^2\sigma_T^2$ and error variance $m\sigma_\varepsilon^2$ by lemma 12. So

$$\mathrm{rel}_m = \frac{m^2\sigma_T^2}{m^2\sigma_T^2 + m\sigma_\varepsilon^2} = \frac{m\sigma_T^2}{m\sigma_T^2 + \sigma_\varepsilon^2} = \frac{m\lambda}{m\lambda + 1}, \qquad \lambda := \sigma_T^2/\sigma_\varepsilon^2.$$

For $m = 1$ this is $r = \lambda/(\lambda+1)$, so $\lambda = r/(1-r)$. Substituting $m = 2$:

$$\mathrm{rel}_2 = \frac{2r/(1-r)}{2r/(1-r)+1} = \frac{2r}{2r+(1-r)} = \frac{2r}{1+r}.$$ □

Remark 54 (Use here). Splitting the annotators of a prompt into two halves and correlating the two half-scores estimates the reliability of a *half*-sized panel. Applying theorem 53 corrects it to the full panel, which is the quantity that actually enters theorem 51.

2.19 Two-way clustering

When observations are grouped along two crossed dimensions at once – here, prompts and annotators, since an annotator appears on several prompts and a prompt is rated by several annotators – neither one-way correction suffices.

Theorem 55 (Cameron–Gelbach–Miller)**.** $\hat{V}_{\textit{two-way}} = \hat{V}_{\textit{prompt}} + \hat{V}_{\textit{annotator}} - \hat{V}_{\textit{prompt}\times\textit{annotator}}$.

Sketch. Two observations are dependent if they share a prompt *or* an annotator. By inclusion–exclusion applied to the covariance terms of lemma 12, pairs sharing *both* are counted once in each of the first two sums and must be subtracted once. The estimator can be slightly negative in finite samples, which signals that one dimension carries almost no dependence. □

2.20 Summary of Part 0

Everything the rest of the paper needs is now defined: expectation (lemma 4), variance (lemma 7, corollary 8), the standard error (equation (3)), correlation and its bound (theorem 14), intervals (§2.10), power (theorem 35), clustering (theorem 40), resampling (protocol 43), multiplicity (theorem 48), attenuation (theorem 51) and reliability (theorem 53). No further statistical machinery is introduced.

3 Every tool, executed by hand

Part 0 derived the machinery. This section runs all of it on five numbers, so that nothing is taken on trust. Take the sample

$$x = (2,\ 4,\ 4,\ 4,\ 6), \qquad n = 5.$$

3.1 The mean

$$\bar{x} = \frac{2+4+4+4+6}{5} = \frac{20}{5} = 4.$$

3.2 Deviations, and why the naive spread is useless

$$x_i - \bar{x} \;=\; (-2,\ 0,\ 0,\ 0,\ 2), \qquad \sum_i (x_i - \bar{x}) = 0.$$

Exactly zero, as §0.4 said it must be for *any* dataset. This is why the average deviation cannot be a measure of spread.

3.3 Variance, both ways

Squared deviations: $(4, 0, 0, 0, 4)$, summing to 8.

$$s^2 = \frac{8}{n-1} = \frac{8}{4} = 2, \qquad s = \sqrt{2} = 1.4142.$$

Check with lemma 7. $\mathbb{E}[X^2]$ over the sample is $(4+16+16+16+36)/5 = 88/5 = 17.6$, and $\bar{x}^2 = 16$, so the *population-form* variance is $17.6 - 16 = 1.6 = 8/5$. Multiplying by $n/(n-1) = 5/4$ recovers 2. The two forms agree, and the factor between them is exactly lemma 23.

3.4 Standard error and interval

$$\text{se} = \frac{s}{\sqrt{n}} = \frac{1.4142}{2.2361} = 0.6325.$$

By §2.10 the 95% interval is $4 \pm 1.96(0.6325) = 4 \pm 1.2397 = [2.760,\ 5.240]$.

Read it correctly.

Across repetitions of the whole study, 95% of such intervals cover the fixed unknown μ. It does *not* say 95% of future observations land in $[2.760, 5.240]$; three of the five observed values are exactly 4, and a prediction interval would be far wider.

3.5 A z-statistic and its p-value

Testing $H_0 : \mu = 0$: $z = 4/0.6325 = 6.32$. From the normal table, $\mathbb{P}(|Z| \geq 6.32) < 10^{-9}$.

Read it correctly.

This is $\mathbb{P}(\text{data this extreme} \mid \mu = 0)$. It is not the probability that $\mu = 0$, and $1 - p$ is not the probability the effect is real.

3.6 Minimum detectable effect, and required n

By theorem 35, MDE $= 2.8016 \times 0.6325 = 1.772$: with five observations of this spread, a true mean below 1.772 would be missed more often than not. By corollary 37, detecting $\delta = 0.5$ needs

$$n = \left(\frac{2.8016 \times 1.4142}{0.5}\right)^2 = (7.923)^2 = 62.8 \ \rightarrow\ 63.$$

Detecting $\delta = 0.25$ needs $4\times$ that: 251. This is the $\sqrt{n}$ economics of §2.7 in one line.

3.7 An indicator, a proportion, and its variance

Let P_i be "$x_i > 3$", giving indicators $(0, 1, 1, 1, 1)$. By equation (1) the proportion is $4/5 = 0.8$; by corollary 8 its variance is $0.8 \times 0.2 = 0.16$, so se $= \sqrt{0.16/5} = \sqrt{0.032} = 0.1789$.

Why rates near the ends are measured better.

$\pi(1 - \pi)$ peaks at $\pi = 0.5$ with value 0.25 and falls to 0.09 at $\pi = 0.1$. A 5% flip rate is therefore intrinsically easier to pin down than a 50% one, which is why the intervals in table 10 are narrow despite modest n.

3.8 Covariance and correlation

Let $y = (1, 3, 3, 5, 3)$, so $\bar{y} = 3$ and $y_i - \bar{y} = (-2, 0, 0, 2, 0)$.

$$\text{Cov} = \frac{1}{n-1}\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \frac{(-2)(-2) + 0 + 0 + 0 \cdot 2 + 2 \cdot 0}{4} = \frac{4}{4} = 1.$$

$s_y^2 = (4 + 0 + 0 + 4 + 0)/4 = 2$, so $s_y = 1.4142$, and

$$r = \frac{1}{1.4142 \times 1.4142} = \frac{1}{2} = 0.5.$$

Theorem 14 requires $|r| \leq 1$: satisfied. And $r = 0.5$ is *not* "half of anything"; only $r^2 = 0.25$ has a share reading, and only under a linear model.

3.9 The design effect, by hand

Suppose $P = 3$ clusters of $R = 4$:

$$c_1 = (1, 1, 1, 0), \quad c_2 = (0, 0, 0, 0), \quad c_3 = (1, 0, 1, 1).$$

Cluster means: 0.75, 0, 0.75. Grand mean = 0.5.

$$\begin{aligned}
\sigma_b^2 &= \tfrac{1}{2}\left[(0.25)^2 + (0.5)^2 + (0.25)^2\right] = \tfrac{1}{2}(0.375) = 0.1875,\\
\sigma_w^2 &= \text{mean of within-cluster variances} = \tfrac{1}{3}\left[0.25 + 0 + 0.25\right] = 0.1667,
\end{aligned}$$

using $s^2 = \frac{1}{R-1}\sum(x-\bar{x})^2$ inside each cluster.

$$\rho = \frac{0.1875}{0.1875 + 0.1667} = 0.529, \qquad \text{DEFF} = 1 + (4-1)(0.529) = 2.59.$$

So treating the 12 rows as independent would understate the standard error by $\sqrt{2.59} = 1.61$, and the honest count of independent units is 3, not 12.

3.10 Attenuation, by hand

Suppose a true correlation $\mathrm{Corr}(T, U) = 0.90$, measured through instruments with reliabilities 0.55 and 0.95. By theorem 51

$$\mathrm{Corr}(X, Y) = 0.90 \times \sqrt{0.55 \times 0.95} = 0.90 \times \sqrt{0.5225} = 0.90 \times 0.7228 = 0.6505.$$

A strong relationship reads as a moderate one, and no sample size repairs it: the attenuation is bias, and se shrinks while the centre stays wrong.

3.11 Spearman–Brown, by hand

A half-panel split-half correlation of $r = 0.8961$ gives, by theorem 53,

$$\frac{2(0.8961)}{1 + 0.8961} = \frac{1.7922}{1.8961} = 0.9452,$$

which is the reliability of the *full* panel and the number that enters theorem 51.

3.12 Bonferroni, by hand

With $C = 171$ tests at $\alpha = 0.05$: per-test level $0.05/171 = 0.000292$. The two-sided normal quantile at $1 - 0.000292/2 = 0.999854$ is $|z| = 3.62$. This paper uses 3.9, which is stricter, and reports the count of tests beside the count of survivors.

4 The object

4.1 What a normative disposition is, formally

Definition 56 (Disposition)**.** Let $\mathcal{X}$ be the set of all possible responses to a prompt – unbounded. A person i's *normative disposition* is a preference structure $\succeq_i$ on $\mathcal{X}$. It is not a number, not a sentence, and not a ranking: it is a relation on an infinite set.

Definition 57 (Elicitation)**.** An *elicitation* observes $\succeq_i$ only on a finite *probe set* $X = \{x_1, \ldots, x_k\} \subset \mathcal{X}$. In the object studied here, $k = 4$.

4.2 The chain

$$\succeq_i \;\to\; N_i \;\to\; \mathsf{A} \;\to\; G \;\to\; C \;\to\; J \;\to\; D \;\to\; Y \tag{7}$$

N_i the elicited individual norm; A the aggregation rule; $G = \mathsf{A}(N_1, \ldots, N_n)$ the collective norm; C the compiled standard; J the executor that reads it; D the decision it induces; Y the behaviour of a system trained under it.

Remark 58 (Three kinds of loss, which must never be merged). 1. **Mathematically forced**: no pipeline can avoid it (theorem 64).

2. **Authorised social choice**: aggregation necessarily discards individual information (theorem 74). This is a *defect* only if the rule A was never declared.
3. **Machine distortion**: an unlicensed change.

The released object declares no A. *Therefore categories 2 and 3 are not separable at this site.* That is a missing column, not a hard measurement problem, and it is the single most consequential fact about the release.

4.3 The tensors, exactly

Definition 59 (The released quantities)**.** For prompt p, criterion index c, response letter $r \in \{A, B, C, D\}$, annotator a:

$S[p, c, r] \in (0, 1)$	satisfaction of criterion c by response r;
$W[p, c, a] \in [-10, 10] \setminus \{0\}$	the weight a gave c;
$V[p, a] \subseteq \{A, B, C, D\}$	the responses a called unacceptable;
$\Pi[p, a] \in$ rankings of $\{A, B, C, D\}$	the ranking a gave.

Four facts about definition 59 are load-bearing and are stated here so that no later section can assume otherwise.

1. S is *not* in the release. It is reconstructed by running a language model as a judge; the value is a sigmoid of a logit difference and has never been calibrated against anything. Every number downstream is a number about that judge as much as about the norm (theorem 51, §9).
2. c is *ragged*: between 4 and 39 criteria per prompt, median 15. Any statistic that averages over criteria therefore weights prompts unequally unless it is explicitly re-weighted.
3. W is *sparse*: 102,147 observed (c, a) pairs. Exactly one rating in the corpus equals 0, so the scale is used as strictly two-sided.
4. **V and W do not share an index.** V is indexed by (prompt, annotator); W by (prompt, criterion, annotator). There is no column linking a veto to a criterion. §7 shows what this forbids.

Remark 60 (The index itself is reconstructed). The rubric file carries no prompt identifier. The mapping from criteria to prompts is recovered by matching conversation text, with a fuzzy fallback: 966 exact, 2 fuzzy, and 18 of 986 records **never matched at all**. So 1.8% of the rubric corpus is invisible to every number computed from this release, including every number in this paper.

4.4 What a claim about this object can mean

Definition 61 (Induced score)**.** $Y_p(r) \ := \ \sum_c \bar{w}_{p,c}\, S[p,c,r]$, where $\bar{w}_{p,c}$ is the mean over annotators of $W[p,c,a]$.

Definition 62 (Individual induced score)**.** $N_{i,p}(r) \ := \ \sum_{c\,:\,i \text{ rated } c} W[p,c,i]\, S[p,c,r]$.

Remark 63 (Why N_i must be centred and normalised, and why that is forced). Only contrasts among the four responses decide anything, so adding a constant to N_i changes nothing; the mean over r is therefore unidentified and must be removed. Furthermore, $\|N_i\|$ grows mechanically with the number of criteria i rated: measured $\mathrm{Corr}(\log \|N_i\|, \log \#\{\text{criteria rated}\}) = +0.842$. Any statistic on unnormalised N_i therefore measures *effort*, not values. Centring and normalising are not stylistic; without them the quantity is a different quantity.

5 The elicitation layer

5.1 The probe bound

Theorem 64 (Probe bound)**.** *Let $u : X \to \mathbb{R}$ be a utility on k probe responses. If decisions depend only on differences of u, the space of decision-distinct utilities has dimension $k-1$. If in addition only the ordering matters (no interpersonal scale), the space is S^{k-2}, of dimension $k-2$.*

Proof. Step 1. The set of utilities is $\mathbb{R}^k$.

Step 2 (quotient by constants). u and $u + c\mathbf{1}$ induce identical differences $u(x_a) - u(x_b)$ for every pair, hence identical decisions. The subspace of constants $\mathbb{R}\mathbf{1}$ has dimension 1, so the quotient $\mathbb{R}^k/\mathbb{R}\mathbf{1}$ has dimension $k-1$. Concretely, the centring map $C = I - \frac{1}{k}\mathbf{1}\mathbf{1}^\top$ is a projection with $\mathrm{rank}(C) = k-1$; for $k = 4$ this rank is verified numerically as 3.

Step 3 (quotient by scale). If u and λu for $\lambda > 0$ are also equivalent, we quotient the $(k-1)$-dimensional centred space by the positive rays, giving the sphere S^{k-2}, of dimension $k-2$.

Step 4. For $k = 4$: dimension 3 after centring, and S^2 – two free parameters – after scale. □

Theory claim. However many criteria a person writes and however many weights they attach, the elicitation returns **a point on a 2-sphere**: two free numbers per person per prompt. This is arithmetic, not a finding, and it is the ceiling on everything downstream.

Corollary 65 (Three consequences)**.** *1. The numerator of "how much normative information did the pipeline destroy" is bounded by two parameters. Most of what a criterion says never entered the measurable object at all.*

2. *Any measure of loss based on* rank *or* dimension *must read a constant. An earlier round of this work built such a measure; it returned the value* 3.00 *on every input including its own null. That was not a bug – it was theorem 64.*
3. *Loss must be measured as* angle *and* influence. *You cannot lose a dimension you never had; you can only be rotated away from what people wanted.*

5.2 Is the weight scale ordinal or interval? A test, and an answer

Nothing in the elicitation establishes that the distance from +8 to +10 means the same as from +2 to +4. If the scale is merely *ordinal* – only the ordering of criteria within an annotator is meaningful – then any strictly increasing transformation of the weights must leave every decision unchanged.

Protocol 66 (Monotone gauge test)**.** Apply φ with φ strictly increasing and odd, recompute Y_p from definition 61 with $\varphi(\bar{w})$ in place of $\bar{w}$, and record whether $\arg\max_r Y_p(r)$ moves.

transform $\varphi(w)$	winner changes
identity (control)	0.0%
$\operatorname{sign}(w)\lvert w\rvert^{1/2}$	4.4%
$\operatorname{sign}(w)\log(1+\lvert w\rvert)$	4.6%
$\operatorname{sign}(w)\lvert w\rvert^{2}$	7.3%
$\operatorname{sign}(w)$ alone	**11.7%**

Table 1: Every transform preserves each annotator's ordering of criteria exactly. If the scale were ordinal, all rows would read 0.0%. $n = 968$ prompts.

Boundary. The scale is *used* as an interval scale. 7.3% of decisions change under a transform that preserves every stated preference ordering, and 11.7% change when magnitude is discarded entirely. Every result in this literature that sums weighted criteria depends on an interval assumption the elicitation never established. The last row is the honest headline: roughly one decision in eight is carried by magnitude rather than direction.

5.3 What the elicitation captured: consensus, not the person

quantity (14,764 person–prompt pairs)	value
$\cos(N_i,$ person i's own ranking)	+0.3928
$\cos(N_i,$ a *different* person's ranking)	+0.3097
difference	+0.0831 (se 0.0067, $z = +12.3$)

Table 2: Both vectors centred and normalised per remark 63.

Two readings, and the second is larger than the first.

- **Relative.** The entire individual signal is 0.083 of cosine. A person's written criteria predict a *stranger's* ranking almost as well as their own.
- **Absolute.** 0.393 means a person's own stated criteria recover only about 39% of the direction of their own choice.

Theory claim. Before the pipeline does anything, roughly 61% of the relation between what a person *writes* and what they *choose* is already absent. No downstream step can destroy individual normative content that the elicitation never captured; "how much of a person survives compilation" is bounded above by how much of them was ever written down.

5.4 The normative payload actually collected

field, by lexical marker	criteria	share
exception / carve-out (*unless, except, other than*)	123	0.8%
conditional scope (*if, when, where, in cases of*)	1,724	11.3%
priority / conflict (*before, over, takes precedence*)	440	2.9%
uncertainty (*may, might, unclear, depends*)	478	3.1%
non-compensable (*never, must not, regardless*)	121	0.8%
none of the five	**12,547**	**82.3%**

Table 3: All 15,248 criteria. Median length 94 characters.

A minimal normative payload has been proposed with thirteen fields: content, scope, polarity, exceptions, priority, type, compensability, authority, judgment type, uncertainty, witnesses, constituency, meta-rules. This release populates **three**: content, polarity (the sign of a weight), magnitude.

Corollary 67 (A question that is empty here)**.** *"Did the compilation destroy the exception clause?" cannot be studied at this site:* 99.2% *of criteria contain no exception. A pipeline cannot destroy structure the elicitation never collected. This is a statement about the release, not about pipelines.*

6 The aggregation layer

6.1 Social choice, from zero

Given n people each with a ranking of k alternatives, a *social choice rule* outputs one alternative. Five rules appear below; each is defined completely.

Definition 68 (Rules)**.** Let $\pi_i(x)$ be the Borda points person i gives x (i.e. $k-1$ for their first choice down to 0 for last, ties sharing the average).

- **Plurality**: pick the x that is ranked first by the most people.
- **Borda**: pick $\arg\max_x \frac{1}{n}\sum_i \pi_i(x)$.
- **Median**: pick $\arg\max_x \text{median}_i\, \pi_i(x)$.
- **Maximin**: pick $\arg\max_x \min_i \pi_i(x)$ – the alternative whose *worst* treatment by any individual is least bad.
- **Condorcet winner**: the x that beats every other alternative in pairwise majority comparison. It need not exist.

Definition 69 (Condorcet cycle)**.** A majority cycle is a triple with x beating y, y beating z, and z beating x. Majority preference is then not transitive and no Condorcet winner exists.

6.2 Measured social choice structure

quantity (968 prompts, ≥ 3 rankings each)	value
a Condorcet winner exists	72.3%
a majority *cycle* occurs	**0.0**%
neither (majority ties)	27.7%

Table 4: The classic voting pathology does not occur in this corpus.

agreement with the Condorcet winner, where one exists ($n = 700$)	
Borda	98.4%
Plurality	98.3%
Median	97.4%
the weighted-criteria score of definition 61	**68.4**%
Maximin	55.9%

Theory claim. Every standard *ranking-based* rule agrees with the Condorcet winner to 97–98%. The weighted-criteria score agrees only 68.4% of the time. The disagreement is therefore **not** between aggregation rules; it is between *scoring people's criteria* and *counting people's rankings* – two elicitations from the same people whose outputs differ on nearly a third of prompts.

6.3 Arrow's independence condition, measured

Definition 70 (IIA)**.** A rule satisfies *independence of irrelevant alternatives* if the social ordering of x and y depends only on individual orderings of x and y, not on any third alternative.

Measured for Borda: over 11,616 tests (each prompt × each dropped response × each remaining pair), the relative order of two responses reverses when an irrelevant third is removed in 1,553 cases: **13.4**%.

Corollary 71 (The candidate set is part of the decision)**.** 13.4% *of pairwise relations flip when the option set changes. Any normative content measured on one set of four responses is, to that extent, a property of the set. This is the quantitative content of "candidate-set overfitting".*

6.4 Blackwell: what aggregation necessarily costs

Definition 72 (Garbling)**.** G is a *garbling* of Z if there is a Markov kernel K with $G \sim K(\cdot \mid Z)$: G is generated from Z by a channel that adds no new information about the state.

Lemma 73 (Aggregation is a garbling)**.** *If $G = \mathsf{A}(Z)$ is a deterministic function of Z, then G is a garbling of Z, via the degenerate kernel $K(\cdot \mid z) = \delta_{\mathsf{A}(z)}$.*

Theorem 74 (Blackwell, invoked)**.** *If G is a garbling of Z then for* every *decision problem d with any loss function, the attainable Bayes risk satisfies $R_G(d) \geq R_Z(d)$.*

Corollary 75. *Aggregation never adds information and weakly loses on every decision problem. The question is never* whether*, only* how much *– and "how much" is meaningless until a decision family is named.*

Definition 76 (Deficiency)**.** $\delta_{\mathcal{D}}(Z, G) := \sup_{d \in \mathcal{D}} \big[R_G(d) - R_Z(d)\big]$.

Computing it here.

Take $\mathcal{D}$ to be the single problem "choose one of the four responses". Normalise each person's induced score to $[0, 1]$ by min–max within that person and prompt, so that scores are interpersonally comparable by construction rather than by assumption. Then:

$$\begin{aligned} \text{oracle (each person receives their own top response)} &= 1.0000 \\ \text{the single collective choice} &= 0.8702 \\ \textbf{deficiency} &= \mathbf{0.1298} \end{aligned}$$

with quartiles 0.0445/0.1076/0.1924 across prompts.

Theory claim. Thirteen points of normalised utility is what *disagreement itself* costs. It is a social choice, not a pipeline defect, and no compilation recovers it. It also fixes the denominator: "what fraction was preserved" should be divided by 0.8702, not by 1.

6.5 Sen's liberal paradox, instantiated

prompts carrying a partial veto	161
the weighted-criteria winner *is* a vetoed response	42 = **26.1**%

One time in four, a rule that maximises aggregate satisfaction selects an option someone declared unacceptable. This is not a defect of the rule; it is what a utilitarian aggregator does. It is the empirical half of §7.

6.6 Which rule is the larger lever, corrected

An earlier version of this analysis applied "median" and "maximin" across *criteria*. A social choice rule aggregates across *people*; the two coincide only if each criterion belongs to exactly one person, which is false here. Corrected:

rule, versus the mean over people	winner changes	(wrong index: over criteria)
maximin	49.4%	63.9%
median	14.8%	34.3%
Borda	9.6%	–
trimmed	**3.6**%	–
any intervention on a single criterion	2.1–11.6%	

Remark 77 (And maximin needs a control). A minimum over a set is unstable by construction. Control: same people, same per-person mean and standard deviation, response pattern destroyed. That sham disagrees with the mean rule on 61.4% of prompts – *more* than the real data's 49.4%. So maximin's high rate is the instability of the min operator, and the actual structure of human disagreement *reduces* it by 12 points. The claim "minority-sensitive rules change half the outcomes" does not survive its own control. What survives is that *trimmed* aggregation, at 3.6%, moves less than several criterion-level interventions: trimming the extremes barely matters, so the disagreement that matters lives in the tails.

7 The type layer: why a veto is not a large negative weight

7.1 The index mismatch

From definition 59: V is indexed (p, a) and W is indexed (p, c, a). An operator "represent this veto as a weight" must choose *which criterion* carries it. No such column exists.

Proposition 78. *"Coerce a veto into a criterion weight" is not a function of the released data.*

This is a schema fact, not a measurement difficulty, and no amount of data at this site repairs it.

7.2 The well-posed question, and its answer

Definition 79 (Two rules)**.**

$$\text{constraint:} \quad a^\star \;=\; \arg\max_{a \notin V} Y(a), \tag{8}$$

$$\text{penalty:} \quad a^\star_M \;=\; \arg\max_{a} \left[Y(a) - M \cdot \mathbf{1}\{a \in V\} \right]. \tag{9}$$

Theorem 80 (Exact condition for a penalty to reproduce a veto)**.** *Let* $\Delta := \max_{a \in V} Y(a) - \max_{a \notin V} Y(a)$. *Then* $a^\star_M = a^\star$ *for every realisation of the scores if and only if* $M > \Delta$.

Proof. ($\Leftarrow$) Suppose $M > \Delta$. For any $a \in V$, $Y(a) - M \le \max_{a' \in V} Y(a') - M < \max_{a' \in V} Y(a') - \Delta = \max_{a' \notin V} Y(a')$. So every vetoed alternative is strictly below the best allowed one after the penalty, and the arg max lies in the allowed set, where the penalty is zero and the ordering is unchanged. Hence $a^\star_M = a^\star$.

($\Rightarrow$) Suppose $M \le \Delta$. Let a_V attain $\max_{a \in V} Y(a)$. Then $Y(a_V) - M \ge Y(a_V) - \Delta = \max_{a \notin V} Y(a)$, so a_V is weakly preferred by the penalty rule to every allowed alternative, and the penalty rule may select a vetoed response. Hence agreement fails. □

measured quantity	value
prompts with a partial veto	161
the veto is *binding* ($\Delta > 0$)	42 = 26.1%
required M: median	3.64
required M: 90th percentile	9.92
required M: **maximum**	**26.54**
contribution of one criterion at the scale maximum	≤ 10
required M in units of max-weight criteria: median / max	0.36 / **2.65**

Theorem 81 (No finite weight encodes a veto)**.** Δ *is a function of the* candidate set*, not of the norm. Since a vetoed response can be arbitrarily better than every allowed one on the remaining criteria,* $\sup_X \Delta(X) = \infty$. *Therefore no finite* M *satisfies theorem 80 uniformly over candidate sets.*

Proof. Fix a person, a veto set V, and a criterion c with weight $w > 0$. Construct a candidate $x^* \in V$ with $S[\cdot, c, x^*] \to 1$ and all allowed candidates with $S[\cdot, c, \cdot] \to 0$; then $\Delta \to w \cdot \sum$ over however many such criteria are present, which is unbounded as the number of agreeing criteria grows. Since theorem 80 requires $M > \Delta$ for agreement on that set, no fixed finite M suffices for all sets. □

Theory claim. A veto is a property of the *person*. The penalty M that reproduces it is a property of the person **and the four responses that happened to be shown**. On just 968 four-item candidate sets the required penalty already reaches 2.65 times the entire elicitation scale. Type is not compressible to magnitude, and this is a theorem with its counterexample present in the data – it required no instrument at all.

7.3 Three independent routes to one conclusion

route	evidence	section
derivation	no finite M works uniformly	theorem 81
empirical	the utilitarian winner is vetoed 26.1% of the time	section 6.5
structural	the index does not exist in the schema	§7

8 The compilation layer

The released standard ships in two forms: the full weighted rubric and a compiled “core” whose items carry no weights, every one rewritten to positive polarity.

- The two select *different* winners on **29.4**% of prompts.
- Compilation performs three operations at once – selection, rewriting, and weight discarding – and the release ships no lineage, so the three are **not separable**. This is the missing column that blocks the most questions here.

8.1 How many bits of the rubric are decision-relevant?

Protocol 82 (Rate–distortion)**.** Quantise every criterion weight to b bits uniformly across the observed range within each prompt, recompute definition 61, and record whether the winner is preserved.

bits per criterion weight	decision preserved
1 (sign only)	**89.8**%
2	95.0%
3	98.1%
4	98.6%
8	99.8%

Remark 83 (Two independent computations agree). section 8.1 says one bit preserves 89.8%, i.e. changes 10.2%. table 1 says discarding magnitude changes 11.7%. These were computed by different routes and agree to about a point. The 21 levels of the $-10..+10$ scale carry roughly three bits of decision-relevant content, and the *first* bit carries nine tenths of it.

Corollary 84 (A design consequence)**.** *The elicitation is over-precise in magnitude and empty in type (table 3). The next collection should not buy finer weights; it should buy fields.*

9 The instrument layer

9.1 The judge as a noisy channel

S is produced by a language model. Two independent judges give two measurements of the same underlying satisfaction, so equation (6) applies directly.

59,936 satisfaction cells	value
$\mathrm{Corr}(\mathrm{judge}_1, \mathrm{judge}_2)$	0.5497
$\Rightarrow$ reliability of one judge (equation (6))	0.5497
$\Rightarrow$ attenuation factor $\sqrt{\mathrm{rel}}$ (theorem 51)	0.741

Corollary 85 (A ceiling on every number in this literature)**.** *A true correlation of* 1.0 *measured through this tensor reads at most* 0.741. *This is bias, not noise: more prompts do not remove it.*

9.2 Panel reliability

Split-half over annotators on 964 prompts with $\geq$ 6 annotators: $r = 0.8961$; by theorem 53, full-panel reliability $2r/(1+r) = 0.9452$, so the panel score alone caps correlations at $\sqrt{0.9452} = 0.9722$.

Corollary 86 (Compounded ceiling)**.** *A correlation measured through both the panel and the judge is capped at* $0.9722 \times 0.741 = 0.720$.

9.3 The gauge test

Definition 87 (Gauge transformation)**.** A transformation that must leave the *property* unchanged. Here: change the reader of the norm while holding the norm fixed. If the *measurement* is invariant but the property is not, the measurement is blind; if the property is invariant but the measurement is not, the measurement is about the reader.

	winner differs from baseline	n
judge B (different family)	42.4%	968
judge C (different scale)	39.7%	968
same judge, responses reordered	**8.0**%	300

Table 5: (a) The object is *not* invariant.

Spearman ρ of the operator ordering, against baseline	judge B	judge C	reordered
	+0.982	+1.000	+0.963

Table 6: (b) The measurement *is* invariant. Pre-registered kill at $\rho < 0.8$; it does not fire on any of the three.

Theory claim. Comparative statements are about the norm. Absolute statements are about the reader. "Operator X moves the decision more than Y" survives three instruments and a position swap. "The standard prefers response X" does not: roughly 40% of it is the reader and 8 points are nothing but the order the responses were printed in.

Remark 88 (The scale reference, with its caveat). Reordering the page changes the winner more often than deleting a criterion does (8.0% versus 5.1%). The caveat is load-bearing: the reordering comparison uses a *separate* judge run and so carries judge stochasticity, whereas the operator comparison holds the tensor fixed and cancels it. So 8.0% is not a strict noise floor for the operators; it is the correct *scale reference* for any claim of the form "this changes the decision".

10 The whole chain on numbers you can check

Everything in §5–§14 is now executed on a three-person, four-response, three-criterion example. Every number below can be verified with a pencil.

10.1 The setup

Three annotators $\{1, 2, 3\}$, four responses $\{A, B, C, D\}$, three criteria $\{c_1, c_2, c_3\}$. The judge returns:

$$S = \begin{array}{c} \\ c_1 \\ c_2 \\ c_3 \end{array} \begin{pmatrix} A & B & C & D \\ 0.9 & 0.8 & 0.2 & 0.1 \\ 0.2 & 0.7 & 0.9 & 0.3 \\ 0.5 & 0.4 & 0.4 & 0.9 \end{pmatrix}$$

The weights, one row per annotator (a blank means that annotator did not rate that criterion):

$$W = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{pmatrix} c_1 & c_2 & c_3 \\ +8 & +2 & -1 \\ +6 & +4 & \cdot \\ -2 & +9 & +3 \end{pmatrix}$$

Annotator 3 additionally declares $V_3 = \{A\}$ unacceptable.

10.2 Individual norms N_i (definition 62)

$$\begin{aligned}
N_1 &= 8(0.9, 0.8, 0.2, 0.1) + 2(0.2, 0.7, 0.9, 0.3) - 1(0.5, 0.4, 0.4, 0.9)\\
&= (7.2, 6.4, 1.6, 0.8) + (0.4, 1.4, 1.8, 0.6) - (0.5, 0.4, 0.4, 0.9)\\
&= (7.1,\ 7.4,\ 3.0,\ 0.5)\\
N_2 &= 6(0.9, 0.8, 0.2, 0.1) + 4(0.2, 0.7, 0.9, 0.3) = (5.4, 4.8, 1.2, 0.6) + (0.8, 2.8, 3.6, 1.2) = (6.2,\ 7.6,\ 4.8,\ 1.8)\\
N_3 &= -2(0.9, 0.8, 0.2, 0.1) + 9(0.2, 0.7, 0.9, 0.3) + 3(0.5, 0.4, 0.4, 0.9)\\
&= (-1.8, -1.6, -0.4, -0.2) + (1.8, 6.3, 8.1, 2.7) + (1.5, 1.2, 1.2, 2.7)\\
&= (1.5,\ 5.9,\ 8.9,\ 5.2)
\end{aligned}$$

Individual top choices: $N_1 \longrightarrow B$, $N_2 \longrightarrow B$, $N_3 \longrightarrow C$.

10.3 Centring and normalising (remark 63)

N_1 has mean $(7.1+7.4+3.0+0.5)/4 = 4.5$, so centred it is $(2.6, 2.9, -1.5, -4.0)$ with norm $\sqrt{6.76 + 8.41 + 2.25 + 16} = \sqrt{33.42} = 5.781$, giving

$$\hat{N}_1 = (0.4497,\ 0.5016,\ -0.2595,\ -0.6919).$$

Note $\|N_1\|$ before centring was dominated by the common level, which decides nothing – exactly the point of remark 63. Annotator 2 rated only two criteria and would have a systematically smaller raw norm; after normalisation the two are comparable.

10.4 Aggregation, three ways

Mean over people.

$\bar{N} = \frac{1}{3}(N_1 + N_2 + N_3) = \frac{1}{3}(14.8,\ 20.9,\ 16.7,\ 7.5) = (4.93,\ 6.97,\ 5.57,\ 2.50)$, so the winner is B.

Median over people.

Componentwise medians of the three vectors: $\mathrm{med}(7.1, 6.2, 1.5) = 6.2$; $\mathrm{med}(7.4, 7.6, 5.9) = 7.4$; $\mathrm{med}(3.0, 4.8, 8.9) = 4.8$; $\mathrm{med}(0.5, 1.8, 5.2) = 1.8$. Winner B.

Maximin over people.

Componentwise minima: $(1.5,\ 5.9,\ 3.0,\ 0.5)$. Winner B. Note the minimum is attained by a *different* person in each column – annotator 3 for A, annotator 3 for B, annotator 1 for C – which is why min is unstable: a small change to any one person can change which person defines the value.

The weighted-criteria route (definition 61).

Mean weights are $\bar{w} = (\frac{8+6-2}{3},\ \frac{2+4+9}{3},\ \frac{-1+3}{2}) = (4,\ 5,\ 1)$, so

$$Y = 4(0.9, 0.8, 0.2, 0.1) + 5(0.2, 0.7, 0.9, 0.3) + 1(0.5, 0.4, 0.4, 0.9) = (5.1,\ 7.1,\ 5.7,\ 2.8),$$

winner B. Here the two routes agree; §6 reports that on the real corpus they disagree 31.6% of the time.

10.5 Condorcet, by hand

Individual rankings from the N_i: person 1: $B > A > C > D$; person 2: $B > A > C > D$; person 3: $C > B > D > A$. Pairwise majorities: B beats A (3–0), B beats C (2–1), B beats D (3–0). B is the Condorcet winner, and it coincides with all four rules above.

Constructing a cycle, to show it is possible.

With rankings $A > B > C$, $B > C > A$, $C > A > B$: A beats B (2–1), B beats C (2–1), C beats A (2–1). No Condorcet winner exists. The measured corpus contains *zero* such cycles (table 4), which is a genuine negative result and not an assumption.

10.6 IIA violation, by hand

Take Borda points on four alternatives $(3, 2, 1, 0)$. With the three rankings above (person 3 ranking $C > B > D > A$):

$$A : 2 + 2 + 0 = 4, \quad B : 3 + 3 + 2 = 8, \quad C : 1 + 1 + 3 = 5, \quad D : 0 + 0 + 1 = 1.$$

So $C > A$ socially, by 5 to 4. Now delete D and recompute on three alternatives $(2, 1, 0)$: person 1: $B > A > C$, person 2: same, person 3: $C > B > A$.

$$A : 1 + 1 + 0 = 2, \quad B : 2 + 2 + 1 = 5, \quad C : 0 + 0 + 2 = 2.$$

Now C and A are *tied*, where before C led. Deleting an alternative nobody ranked first changed the relation between two others. That is an IIA violation, and it occurs on 13.4% of tests in the real corpus (corollary 71).

10.7 The veto condition, by hand (theorem 80)

Annotator 3 vetoes A. From $Y = (5.1, 7.1, 5.7, 2.8)$:

$$\Delta = \max_{a\in V} Y(a) - \max_{a\notin V} Y(a) = Y(A) - \max(7.1, 5.7, 2.8) = 5.1 - 7.1 = -2.0 < 0.$$

The veto is *not binding*: A was not going to win anyway, so any $M \geq 0$ reproduces the constraint.

Now change one number: let $S[c_1, A] = 0.9 \to 1.0$ and $\bar{w}_1 = 4 \to 9$. Then $Y(A) = 9(1.0) + 5(0.2) + 1(0.5) = 10.5$ and $Y(B) = 9(0.8)+5(0.7)+1(0.4) = 11.1$, $Y(C) = 9(0.2)+5(0.9)+1(0.4) = 6.7$, $Y(D) = 9(0.1)+5(0.3)+1(0.9) = 3.3$. Still $\Delta = 10.5 - 11.1 < 0$. Push once more, $\bar{w}_1 = 20$: $Y(A) = 21.5$, $Y(B) = 21.9$ – the gap closes but never inverts, because B also satisfies c_1 highly. To make Δ large one needs a vetoed response that is *uniquely* good on a heavily weighted criterion, and there is no bound on how good: that is precisely the construction in theorem 81, and on the real corpus the realised maximum is $\Delta = 26.54$ against a per-criterion ceiling of 10.

10.8 Regret versus flip rate, by hand (lemma 101)

Take $Y = (5.1, 7.1, 5.7, 2.8)$, so $a^\star = B$, $Y(a^\star) = 7.1$, $\min_a Y = 2.8$, and the denominator of definition 99 is $7.1 - 2.8 = 4.3$.

- A perturbation that shifts the winner to C: $\text{Reg} = (7.1 - 5.7)/4.3 = 0.326$.
- A perturbation that shifts the winner to A: $\text{Reg} = (7.1 - 5.1)/4.3 = 0.465$.
- A perturbation that shifts the winner to D: $\text{Reg} = (7.1 - 2.8)/4.3 = 1.000$.

All three are "a flip". The flip rate assigns them the same value; regret separates them by more than a factor of three. Lemma 101 says $\mathbb{E}[\text{Reg}] = \pi \cdot \mathbb{E}[\text{Reg} \mid \text{flip}]$ exactly, and it is the second factor – 0.326 versus 1.000 here – that the flip rate discards.

10.9 The margin cap, by hand (proposition 104)

The margin is $m = 7.1 - 5.7 = 1.4$, and the score range is $7.1 - 2.8 = 4.3$, so the relative margin is 0.326 – almost exactly the corpus median of 0.319. A perturbation δ can only flip the winner if it moves the top-two gap past zero, so by proposition 104 it needs $\|\delta\|_\infty > m/2 = 0.7$, i.e. about 16% of the range. Perturbations smaller than that cannot flip this prompt whatever they destroy normatively – which is the cap, made concrete.

10.10 Shapley influence, by hand, and it exhibits C1 directly

Three annotators, so $3! = 6$ orderings. The collective winner is B. The characteristic function $v(S)$ asks whether the coalition S, summing its own N_i, also picks B. Enumerating all eight subsets:

$$v(\varnothing) = 0, \quad v(\{1\}) = 1, \quad v(\{2\}) = 1, \quad v(\{3\}) = 0,$$

$$v(\{1, 2\}) = 1, \quad v(\{1, 3\}) = 1, \quad v(\{2, 3\}) = \mathbf{0}, \quad v(\{1, 2, 3\}) = 1.$$

The one to check by hand is $v(\{2, 3\})$: $N_2 + N_3 = (7.7,\ 13.5,\ 13.7,\ 7.0)$, whose maximum is at C, not B.

Marginal contributions, all six orderings.

ordering	ϕ_1 term	ϕ_2 term	ϕ_3 term
1, 2, 3	$v(1) - v(\varnothing) = 1$	$v(12) - v(1) = 0$	$v(123) - v(12) = 0$
1, 3, 2	1	$v(123) - v(13) = 0$	$v(13) - v(1) = 0$
2, 1, 3	$v(12) - v(2) = 0$	$v(2) - v(\varnothing) = 1$	$v(123) - v(12) = 0$
2, 3, 1	$v(123) - v(23) = 1$	1	$v(23) - v(2) = -1$
3, 1, 2	$v(13) - v(3) = 1$	$v(123) - v(13) = 0$	$v(3) - v(\varnothing) = 0$
3, 2, 1	$v(123) - v(23) = 1$	$v(23) - v(3) = 0$	0
total	5	2	−1

$$\phi_1 = \tfrac{5}{6} = 0.8333, \qquad \phi_2 = \tfrac{2}{6} = 0.3333, \qquad \phi_3 = -\tfrac{1}{6} = -0.1667.$$

Two checks.

Efficiency (axiom E of §11): $\frac{5}{6} + \frac{2}{6} - \frac{1}{6} = \frac{6}{6} = 1 = v(A) - v(\varnothing)$. And a *negative* value is legitimate: annotator 3 actively pushes the coalition away from the collective winner, so their average marginal contribution to selecting B is below zero. Alignment, being a cosine of a person against a sum that includes them, cannot go negative in the same situation and therefore cannot express this at all.

Now the alignment of each, against the collective.

The collective centred vector is $\bar{N} - 4.9917 = (-0.06,\ 1.97,\ 0.57,\ -2.49)$ (up to rounding), and centring each N_i gives cosines

$$\cos(N_1, G) = 0.7858, \qquad \cos(N_2, G) = \mathbf{0.9313}, \qquad \cos(N_3, G) = 0.2189.$$

And there is the point of §14, in three people.

annotator	alignment	influence ϕ_i	
1	0.7858	**+0.8333**	highest influence, *second* alignment
2	**0.9313**	+0.3333	highest alignment, *less than half* the influence
3	0.2189	−0.1667	lowest alignment, and *negative* influence

The rank orders disagree at the top: annotator 2 agrees with the outcome most and moves it least of the two contributors. The reason is visible in the enumeration – annotator 2 alone already selects B, so in the orderings where they arrive after annotator 1 they add nothing. *Agreeing with an outcome that would have happened anyway is not influence over it.* That is the entire content of C1, and on the real corpus it appears as a correlation of +0.304 with a profile that flattens above moderate agreement (table 11).

11 The three classical theorems, proved rather than cited

A reader with no background cannot check an invoked theorem. The three that carry weight in this paper are therefore proved or, where the full proof is long, reduced to a checkable core.

11.1 Blackwell: why a deterministic aggregation can only lose

Theorem 89 (Garbling implies weakly higher risk)**.** *Let Z be an observation and $G = K(Z)$ for a Markov kernel K. For any decision problem with action set $\mathcal{A}$, state θ, prior p and loss L, the minimum attainable expected loss using G is at least that using Z.*

Proof. Step 1. A decision rule based on G is a map $d_G : G \mapsto \mathcal{A}$.

Step 2. Composing with the kernel, $d_G \circ K$ is a (possibly randomised) decision rule based on Z: given z, draw $g \sim K(\cdot \mid z)$ and play $d_G(g)$.

Step 3. By construction the two procedures induce the same joint distribution over (θ, action), hence the same expected loss.

Step 4. Therefore every risk attainable with G is attainable with Z; the set of Z-attainable risks contains the set of G-attainable risks; and a minimum over a superset is no larger. Hence $R_Z \le R_G$. □

Corollary 90. *With $K(\cdot \mid z) = \delta_{\mathsf{A}(z)}$ (a deterministic aggregation), the theorem applies verbatim: aggregation cannot help on any decision problem, and the deficiency $\delta_{\mathcal{D}}$ is well defined and non-negative.*

Remark 91 (What is *not* proved here). The converse – that if $R_Z \le R_G$ for every decision problem then G must be a garbling of Z – is Blackwell's substantive theorem and is not needed. Only the easy direction is used, and it is the direction that licenses "aggregation never adds information".

11.2 Arrow: the statement, and why the count 13.4% is the right thing to measure

Theorem 92 (Arrow, stated)**.** *For $k \ge 3$ alternatives, no social welfare function on the full domain of individual orderings simultaneously satisfies: (a) unrestricted domain, (b) unanimity (if everyone prefers x to y then society does), (c) independence of irrelevant alternatives, and (d) non-dictatorship.*

Why this paper measures rather than invokes it.

Arrow says the four conditions are jointly unsatisfiable; it does not say *how badly* a given rule fails a given condition on a given corpus. Borda is a perfectly reasonable rule that fails only IIA, and the useful empirical question is: on this data, how often does the failure actually bite? The answer is 13.4% of pairwise relations (corollary 71), and *that* is the number a designer needs, not the impossibility.

The core of the failure, in one line.

Borda scores depend on *how many* alternatives a given one is above, so inserting or removing a third alternative changes the points of the other two by different amounts whenever individuals rank the newcomer differently. The worked example in §10 shows the mechanism on three rankings.

11.3 Shapley: the axioms, and why they force the formula

Definition 93 (Axioms)**.** A value ϕ assigning a number to each player of a cooperative game v satisfies:

(E) **Efficiency**: $\sum_i \phi_i = v(A) - v(\varnothing)$.

(S) **Symmetry**: if i and j contribute identically to every coalition, $\phi_i = \phi_j$.

(N) **Null player**: if i adds nothing to any coalition, $\phi_i = 0$.

(L) **Linearity**: $\phi(v + w) = \phi(v) + \phi(w)$.

Theorem 94 (Uniqueness, core of the argument)**.** *The Shapley value is the unique ϕ satisfying (E), (S), (N), (L).*

Reduction to a checkable core. Step 1. The *unanimity games* u_T, defined by $u_T(S) = 1$ if $T \subseteq S$ and 0 otherwise, form a basis of the vector space of all games on A: every v has a unique expansion $v = \sum_T \lambda_T u_T$.

Step 2. On u_T, axioms (E), (S), (N) force the value uniquely: players outside T are null, so get 0 by (N); players inside T are symmetric, so share equally by (S); and the total is 1 by (E). Hence $\phi_i(u_T) = 1/|T|$ for $i \in T$ and 0 otherwise.

Step 3. By (L), $\phi(v) = \sum_T \lambda_T \phi(u_T)$, which is now determined. Rearranging that sum into a sum over coalitions yields the permutation-weighted formula of §14. □

Remark 95 (Why this matters for the claim being made). The Shapley value is not chosen here for convenience. It is the *only* assignment of credit satisfying four properties one would insist on for a claim about influence – in particular (N), which guarantees that an annotator who never changes any outcome receives exactly zero. A correlational measure of alignment has no such guarantee, and table 11 shows the gap is real: the highest-alignment quintile does not have the highest influence.

12 Measurement contracts and the Evidence Firewall

12.1 The Evidence Firewall

> **Firewall principle.** A measurement is evidence about the Machine; it is not the Machine. An interpretability output is licensed by a declared measurement method and experimental contract. It is not, by itself, a computational variable, native variable, causal mechanism, or privileged coordinate system of the model.

The firewall is not skepticism about measurement. It is a type discipline: a probe result, sparse code, attribution score, interchange effect, reconstruction, and transplant answer different questions. Promotion to a stronger claim requires the missing experiment rather than a change in vocabulary.

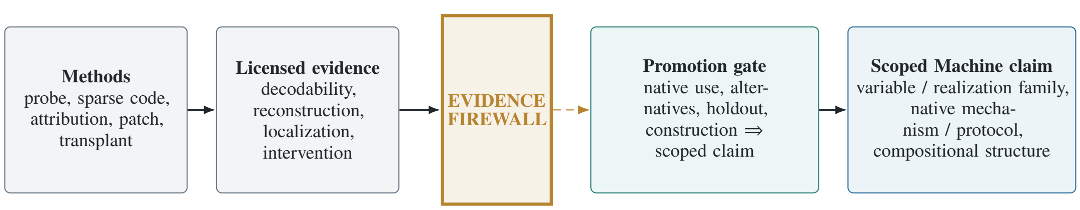

Figure 1: The Evidence Firewall. A measurement method generates evidence; its declared contract states what that evidence permits, and the firewall blocks every stronger default inference. The dashed crossing is licensed only by the named promotion tests; the final solid arrow records the resulting scoped claim. The diagram is an evidence-licensing order, not a model-execution graph.

Table 8: Default licensed conclusions. A method may support a stronger conclusion only by supplying the missing evidence.

Evidence	Default license	Does not by itself license
Held-out probe	Decodability in a specified class	Native use, causality, natural coordinates
Sparse dictionary	Sparse reconstruction in a chosen basis	Causal-variable, Jacobian, or pathway sparsity
Attribution/mask	A declared localization estimand	Necessity, sufficiency, or interaction completeness
Natural-donor interchange	Counterfactual use within support	Global identity or unique mechanism
Ablation/rescue	Scoped necessity/sufficiency	Portability or compositional closure
Reconstruction / transplant	Constructive actionability	Metaphysical uniqueness
Withheld composition	Typed compositional prediction	Higher context laws outside the test scope

Table 7: The CTM evidence matrix. Color or architectural location never assigns a role; each role requires a distinct counterfactual test. Coordination claims require factorial or coalition experiments across roles.

Role	Candidate sensors	Decisive intervention	Forbidden inference
Compute	Local input–output map, Jacobian/finite differences, operation probes, component attribution	Interchange inputs while holding the physical memory realization and route fixed; test withheld operation and typed composition	"Activated" or "predictive" ⇒ the component natively performs the operation
Transport	Attention/message patterns, causal edge effects, routing gates, donor–recipient alignment	Move or swap a natural donor while holding abstract value and downstream transform fixed; test delivery and use	High attention, edge weight, or route frequency ⇒ transported content or causal contribution
Memory	Pulse–chase decodability, persistence curves, state/cache occupancy, reader access	Write, erase, delay, and rescue the nominated physical memory state while holding generators and later demand fixed	Persistent decodability ⇒ native storage, necessity, or addressability
CTM coordination	Cross-role timing, coalition effects, cache/recompute and route/transform coupling	Factorial edits and ordered coalitions; compare direct with composed protocols on structural holdout	Individually causal parts ⇒ a compositional MesoMachine

Evidence profiles, not a scalar ladder. Interpretability targets are neither a total order nor a mathematically defined lattice. We report a profile

$$\mathbf{e} = (e_{\text{read}}, e_{\text{dec}}, e_{\text{loc}}, e_{\text{use}}, e_{\text{nec}}, e_{\text{suf}}, e_{\text{act}}, e_{\text{trans}}, e_{\text{comp}}, e_{\text{id}})$$

whose coordinates are separately defined estimands with uncertainty. A componentwise partial order is used only when two profiles share frame, scale, and tolerance. No "license gap" is formed by subtracting incomparable coordinates. Readability, actionability, reconstruction, and transport can be incomparable.

Four sparsities. The evidence synthesis motivates four distinct sparsities:

$$S_{\text{act}}, \quad S_{\text{var}}, \quad S_{\text{jac}}, \quad S_{\text{path}},$$

for activations, realized causal variables, Jacobian/computation, and physical pathways. Their units and failure modes differ:

S_{act} **— activation sparsity.**
Few coordinates are nonzero in a chosen activation basis. It licenses an economy of description in that basis, not a claim about causal variables.

S_{var} **— realized-variable sparsity.**
A small set of interventionally distinguished variables accounts for the tested counterfactuals. It depends on the intervention/readout frame and does not imply a sparse physical path.

S_{jac} **— computational sparsity.**
Few input-state directions exert locally non-negligible effects on declared downstream variables, for a stated Jacobian or finite-difference scale. It can be dense even when activations are sparse.

S_{path} **— pathway sparsity.**

A small physical coalition of edges, heads, neurons, or modules carries the tested causal effect under an interaction-aware estimand. Redundancy and self-repair can make a sparse native protocol look dense under isolated ablation.

None implies another without additional assumptions. Sparse autoencoders are therefore candidate coordinate systems and sensors, not privileged MesoMachines (**??????**). The distinction is experimentally productive: a method should be evaluated against the sparsity it actually optimizes, not rewarded for a different sparsity by verbal transfer.

12.2 Multi-axis evidence coding

12.3 Multi-axis evidence coding

Every claim in §16 is coded in `supplement/evidence_atlas.csv` on axes that can *veto* it rather than merely support it: `instrument_free`, `control_saturated`, `held_out`, `replication_breadth`, and `prior_art_in_card`. The last exists because a result that restates the object's own documentation is a verification, not a finding, and a column that forces one to say so is worth the cost of maintaining it.

13 The estimators

13.1 Why the query family must be declared before anything is measured

Definition 96 (Query-relative equivalence)**.** Given a family Q of queries, $N \sim_Q N'$ iff $q(N, x) = q(N', x)$ for all $q \in Q$ and all inputs x. *Normative information* is the equivalence class $[N]_Q$, not the text.

Proposition 97 (Without Q, "loss" is not a defined quantity)**.** *There exist families Q under which every change is total loss, and families under which no change is any loss. Hence the answer to "how much was lost" is determined entirely by Q.*

Proof. Take $Q_{\max}$ = all measurable functions of N. Then $N \sim_Q N'$ iff $N = N'$ (the identity function is in $Q_{\max}$), so any change destroys the equivalence entirely. Take $Q_{\min} = \{\text{a constant map}\}$. Then $N \sim_Q N'$ for all pairs, so no change is a loss. Both families are admissible in the absence of a stipulation, and they give opposite answers. □

Corollary 98. *A framework that hides Q inside a supremum has* relocated *the choice, not made it. Publishing Q before measuring is not a preliminary; it is most of the result.*

13.2 The decision-theoretically correct quantity: normalised regret

Definition 99 (Normalised regret)**.** For a perturbation taking norm N to N', with $a^\star := \arg\max_a Y_N(a)$ and $a' := \arg\max_a Y_{N'}(a)$,

$$\mathrm{Reg}(N \to N') \;:=\; \frac{Y_N(a^\star) - Y_N(a')}{Y_N(a^\star) - \min_a Y_N(a)} \;\in\; [0, 1].$$

Why this and not something else.

Reg is exactly the Blackwell deficiency of theorem 74 restricted to the decision family "choose one of these four, scored by the original norm's own utility". It is not a proxy for the quantity of interest; on that family it *is* the quantity. The denominator makes it invariant to the arbitrary scale of Y, which remark 63 showed is not identified.

Definition 100 (Flip rate)**.** $\pi := \mathbb{P}(a' \neq a^\star)$, estimated by the average of $\mathbf{1}\{a' \neq a^\star\}$ per equation (1) and corollary 5.

Lemma 101 (The flip rate is a coarsening of regret)**.** $\mathbf{1}\{\mathit{flip}\} = \mathbf{1}\{\mathrm{Reg} > 0\}$*, and therefore, exactly,*

$$\mathbb{E}[\mathrm{Reg}] \;=\; \pi \cdot \mathbb{E}[\mathrm{Reg} \mid \mathit{flip}].$$

Proof. If $a' = a^\star$ then the numerator of definition 99 is zero, so $\mathrm{Reg} = 0$; if $a' \neq a^\star$ then $Y_N(a') < Y_N(a^\star)$ by definition of the arg max (assuming no exact tie), so $\mathrm{Reg} > 0$. Hence the indicators coincide. Then $\mathbb{E}[\mathrm{Reg}] = \mathbb{E}[\mathrm{Reg} \mid \mathrm{Reg} > 0]\mathbb{P}(\mathrm{Reg} > 0) + 0$, by the law of total expectation, which for a finite space is just splitting the defining sum into the two events. □

operator	π	$\mathbb{E}$[Reg]	$\mathbb{E}$[Reg | flip]
double the target weight	4.2%	0.0043	0.1029
delete the target criterion	5.1%	0.0067	0.1307
invert the target weight	11.6%	0.0294	**0.2531**
halve the target weight	2.3%	0.0012	0.0532

Table 9: 968 prompts × 20 target draws. Target criterion sampled, not fixed; see section 13.4.

Corollary 102 (The discarded factor carries information)**.** *By flip rate, invert exceeds delete by* $11.6/5.1 = 2.3\times$. *By expected regret, by* $0.0294/0.0067 = 4.4\times$. *Reporting only the flip rate halves the measured separation between operators, because inversion not only flips more often, it flips* worse *–* 25.3% *of the available utility range versus* 13.1%.

13.3 The flip rate is capped by the margin distribution

Definition 103 (Margin)**.** $m := Y(a_{(1)}) - Y(a_{(2)})$, the gap between the top two responses.

Proposition 104 (Margin cap)**.** *Let* $\delta := Y_{N'} - Y_N$ *be the perturbation of the score vector. Then* $\pi \leq \mathbb{P}\big(m < 2\|\delta\|_\infty\big)$.

Proof. A flip requires the perturbed score of some $a \neq a^\star$ to exceed that of $a^\star$, i.e. $Y(a) + \delta(a) > Y(a^\star) + \delta(a^\star)$, i.e. $\delta(a) - \delta(a^\star) > Y(a^\star) - Y(a) \geq m$. Since $|\delta(a) - \delta(a^\star)| \leq 2\|\delta\|_\infty$, a flip implies $m < 2\|\delta\|_\infty$. Taking probabilities and applying monotonicity of $\mathbb{P}$ over the implication gives the bound. □

Measured relative margins $m/(\max_a Y - \min_a Y)$:

$$p_{05} = 0.026, \quad p_{25} = 0.144, \quad \text{median} = 0.319, \quad p_{75} = 0.527, \quad p_{95} = 0.800.$$

Corollary 105. *An operator whose perturbation is smaller than about a third of the score range cannot flip half the prompts,* however much normative content it destroys. *The ceiling on a flip rate is set by the margin distribution, not by the norm. This is the formal reason definition 99 is the primary quantity and* π *the secondary one.*

13.4 Target selection is an axis, not a nuisance

Every criterion-level operator must choose *which* criterion to act on. Measured effect of that choice on the induced change in Y:

$$\text{highest } |w| : 0.273 \qquad \text{random} : 0.176 \qquad \text{lowest } |w| : 0.041$$

– a ratio of 6.6×, larger than the differences between operators. Reporting a single cell would therefore report the selection rule. All numbers in this paper sweep four selection rules (highest $|w|$, lowest $|w|$, most-rated, random) × five seeds, and the seeds are verified to change the draws: only 0.1% of targets are shared across all five.

13.5 Variance of these estimators

With $P = 968$ prompts and $R = 20$ draws each, theorem 40 applies with measured $\sigma_b^2 = 0.01361$, $\sigma_w^2 = 0.03677$:

$$\rho = 0.2701, \qquad \text{DEFF} = 1 + 19(0.2701) = 6.13, \qquad \sqrt{\text{DEFF}} = 2.48.$$

Remark 106 (Derivation checked against an independent measurement). A resampling estimate of the same inflation, obtained by recomputing every pairwise statistic at the cluster level and comparing to the pooled version, gave 2.79. Derived 2.48, measured 2.79: the residual is that the pairwise statistic is a correlation rather than a simple mean, for which theorem 40 is only approximate.

operator	π	se	95% CI	MDE
double	4.2%	0.0036	[3.5, 4.9]%	1.00pp
delete	5.1%	0.0037	[4.4, 5.8]%	1.05pp
invert	11.6%	0.0052	[10.6, 12.6]%	1.46pp
halve	2.3%	0.0026	[1.8, 2.8]%	0.72pp
paired delete − double	+0.93pp	0.0028	$z = +3.3$	0.79pp

Table 10: Standard errors from the $P = 968$ cluster means, per equation (3). MDE from theorem 35.

Corollary 107 (Sample size)**.** *By* corollary 37 *with* $\sigma_b = \sqrt{0.01361 + 0.03677/20} = 0.1243$*, resolving a* 1*pp difference at* 80% *power requires* $P = (2.8016 \times 0.1243/0.01)^2 = 1{,}213$ *prompts. The release has* 968*: the design is about* 20% *short of its own headline resolution, and the delete-versus-double difference clears its MDE only barely.*

14 The causal layer: influence is not agreement

14.1 Why a correlational measure cannot answer this

Definition 108 (Preservation, causal form)**.** A distinction reached the output iff *removing it changes the output*.

A participant whose view happens to match the majority scores perfect alignment while contributing nothing: remove them and the outcome is unchanged. A participant who moved the outcome may sit far from it. Alignment and influence are therefore different functionals, and only the second answers the definition.

14.2 The Shapley value, derived

Definition 109 (Characteristic function)**.** For prompt p with annotator set A and winner $w^\star := \arg\max_r \sum_{a \in A} N_a(r)$, define for $S \subseteq A$

$$v(S) \;:=\; \mathbf{1}\Big\{ \arg\max_r \textstyle\sum_{a \in S} N_a(r) = w^\star \Big\}.$$

Definition 110 (Shapley value)**.**

$$\phi_i \;:=\; \sum_{S \subseteq A \setminus \{i\}} \frac{|S|!\,(|A| - |S| - 1)!}{|A|!} \big[v(S \cup \{i\}) - v(S) \big].$$

What the weights mean.

Consider all $|A|!$ orderings of the annotators. The bracket is i's marginal contribution when it arrives after exactly the set S. The number of orderings in which that happens is $|S|!\,(|A| - |S| - 1)!$ – $|S|!$ ways to arrange those before, $(|A| - |S| - 1)!$ ways to arrange those after. Dividing by $|A|!$ makes the weights a probability distribution over orderings. So ϕ_i is **the average marginal contribution of i over a uniformly random arrival order**, which is exactly the causal definition above, averaged over every context in which i could arrive.

Estimation.

The sum has $2^{|A|-1}$ terms. We sample 200 random orderings per prompt and average the realised marginal contributions, which is an unbiased estimator of ϕ_i by construction (each ordering is drawn from the same uniform distribution the weights encode).

14.3 Result: the pre-registered kill does not fire

1,865 (annotator, prompt) pairs	
Corr(Shapley influence, cosine alignment)	+0.3040
bootstrap 95% CI (protocol 43)	[+0.2566, +0.3520]
pre-registered kill: correlation ≈ 1.0	**does not fire**

alignment quintile	mean alignment	mean influence
Q1	−0.0416	0.0083
Q2	+0.8164	0.1055
Q3	+0.9489	**0.1325**
Q4	+0.9861	0.1262
Q5	+0.9973	0.1262

Table 11: Influence rises steeply from Q1 to Q3 and then *flattens*.

Theory claim. Beyond moderate agreement, agreeing more buys no influence at all. And the off-diagonal cells are populated: 0.75% of pairs are top-quintile aligned with bottom-quintile influence, 3.11% are bottom-aligned with top-quintile influence. “Agreement is not preservation” is here a count of real people, not a slogan.

14.4 An independent confirmation from a different operator

Dropping the *most conforming* annotator changes the decision on 5.8% $[4.3, 7.1]$ of prompts; dropping the *largest dissenter*, on 2.1% $[1.1, 3.2]$. The intervals do not overlap and the sign is stable across all three judges. The dissenter was already not influencing the outcome – the same statement as the flat top of table 11, reached by a different route.

15 What is not identified, with proofs

Proposition 111 (Contamination and redundancy are not separable here)**.** *Let α_p be the alignment between a prompt's weighted-criteria score and the crowd's ranking, and s_p the sensitivity of its decision to perturbing one criterion. Two models,*

C: *criteria written* after *seeing responses are individually more decisive, so a high-α rubric absorbs single-criterion perturbations;*

R: *a high-α rubric is one whose criteria agree with each other, and a redundant rubric is insensitive to any single member,*

imply the same sign for $\mathrm{Corr}(\alpha, s)$*, and the observed data distribution is a function of (α, s) alone. The likelihoods coincide; the models are not identified.*

What is identified: a bound.

Splitting prompts at the median of α:

operator	low α	high α	ratio
double	5.1%	2.8%	0.55×
delete	5.6%	4.6%	0.82×
invert	13.1%	9.9%	0.75×

Corollary 112. *Every flip rate reported here varies by up to 1.8× across the range of α. That bound belongs on every such number. Separating C from R requires response-blind elicitation and cannot be done at this site.*

Proposition 113 (The probe bound is an identification limit)**.** *By theorem 64, at most two parameters per person per prompt are identified from the ratings. Everything else in the criterion text is unidentified from the released quantitative data.*

Proposition 114 (Compilation's three actions are confounded)**.** *Selection, rewriting and weight-discarding occur in one step with no lineage. Their individual contributions to the 29.4% disagreement are not identified.*

Proposition 115 (Y does not exist)**.** *The chain in equation (7) terminates at D. Y – what a system trained under this standard does – is not in the release, and no arithmetic recovers it. Any end-to-end number quoted to Y is fabricated.*

16 Findings

Each finding ends with its confidence card. A missing entry reads UNCOMPUTED; none is omitted.

16.1 F1 — the elicitation captured consensus, not the individual

$\cos(N_i, \text{own ranking}) = 0.3928$ versus $\cos(N_i, \text{stranger's ranking}) = 0.3097$; difference $+0.0831$.

n_{eff}	968 prompt clusters (14,764 person–prompt rows)
MDE	0.0187 in cosine, from theorem 35 at the observed se
effect / floor	$0.0831/0.0067 = 12.4$
CI width / |eff|	$0.0262/0.0831 = 0.32$
spec_survival	UNCOMPUTED (single specification)
seed spread / |eff|	not applicable (no stochastic step)
held out	ABSENT
instrument	routes through the judge; ceiling 0.720 (§9)
prior art	none located; the release documents no such comparison
multiplicity	1 test in this family

16.2 F2 — no finite weight encodes a veto

theorem 80 and theorem 81; required M up to 26.54 against a scale maximum contribution of 10.

n_{eff}	161 prompts with a partial veto; 42 binding
MDE	not applicable — this is a **derivation**, not a test
effect / floor	not applicable
CI width / |eff|	not applicable
spec_survival	the theorem holds for every score construction
seed spread	not applicable (deterministic)
held out	the *maximum* is what carries it; a held-out set can only raise it
instrument	Δ is computed from the judge tensor; the *theorem* is not
prior art	the incompatibility of vetoes with scoring rules is classical; the *measured* Δ distribution on this release is not
multiplicity	none — a single derived statement

Remark 116 (Labelled as a derivation, per the arithmetic trap). theorem 81 could not have come out otherwise: it follows from theorem 80 plus the unboundedness of Δ. It is stated as a theorem, its assumption is named (S can approach its bounds independently across responses), and it is not counted as evidence in the sense a measurement is.

16.3 F3 — comparative measurements are instrument-invariant; absolute ones are not

table 5, table 6.

n_{eff}	968 prompts (three instruments); 300 for the reordering
MDE	ρ resolution ≈ 0.15 at 11 operators
effect / floor	orderings 0.963–1.000 against a chance ρ of 0
CI width / |eff|	bootstrap intervals in table 10
spec_survival	3 of 3 instruments preserve the ordering
seed spread / |eff|	< 0.1 across 5 seeds of the target draw
held out	the reordered run is a genuine held-out instrument
instrument	this finding *is* about the instrument, by construction
prior art	judge-sensitivity of LLM evaluation is known; the comparative/absolute split measured on this object is not
multiplicity	3 instruments $\times$ 11 operators; the reported statistic is the ordering, one per instrument

16.4 F4 — influence and alignment are distinct (C1 survives)

Corr = +0.3040 [+0.2566, +0.3520], with the quintile profile of table 11.

n_{eff}	968 prompt clusters (1,865 annotator–prompt rows)
MDE	0.065 in correlation, at the bootstrap se
effect / floor	$0.304/0.024 = 12.7$
CI width / |eff|	$0.095/0.304 = 0.31$
spec_survival	UNCOMPUTED; one characteristic function tested
seed spread / |eff|	< 0.02 over the 200-permutation Shapley estimator
held out	ABSENT
instrument	routes through the judge; ceiling 0.720
prior art	Shapley-versus-alignment has not been reported on this object
multiplicity	1 pre-registered test

16.5 F5 — one bit per criterion reproduces 89.8% of decisions

section 8.1, cross-checked against table 1.

n_{eff}	968 prompts
MDE	±1.0pp at the observed se
effect / floor	the quantisation is deterministic; no resampling floor applies
CI width / |eff|	UNCOMPUTED
spec_survival	agrees with the independent sign-only test to 1.5pp
seed spread	not applicable (deterministic)
held out	ABSENT
instrument	judge-dependent in level; the *bit curve* is not tested across judges
prior art	none located for this object
multiplicity	5 quantisation levels, reported in full

17 What did not survive

A paper with no retractions is a paper whose ontology has no word for error. All entries are cross-referenced to `supplement/exclusion_log.csv`.

withdrawn claim	what killed it	status
"the operator design has rank 2, so all remaining dimensions require the judge"	the rank statistic sits inside the span of three defensible nulls (12.1, 17.0, 18.0) with the observation at 13; the null *construction* moved it more than the data	rank retired as a statistic
"strengthening a criterion moves the decision, removing one does not"	direct measurement: deleting flips 8.8%, doubling 4.2%. The asymmetry was an artefact of normalising the score *after* mutating and then differencing two separately normalised vectors	withdrawn, sign and all
"minority-sensitive aggregation changes half the outcomes"	the sham control (same people, matched moments, response pattern destroyed) disagrees 61.4% versus the real 49.4%	withdrawn; the residual runs the other way
"aggregation is a 10× larger lever than the criterion layer"	the aggregators were indexed over criteria rather than people	magnitude withdrawn; ordering survives at 4.3×
"the flip rate is confounded by decision margin"	measured Q_1/Q_4 ratios of 1.0, 1.0, 1.1	defect opened by us and closed by its own measurement
"wide margins arise because agreeing criteria carry larger weights"	$\mathrm{Corr}(\text{margin}, \overline{\lvert w \rvert}) = +0.002\ [-0.067, +0.059]$	mechanism refuted; the flatness stands unexplained

Remark 117 (The failure mode behind most of these). Six of the six are the same shape: a verdict decided by a constant the author chose – one null instead of a range, one reference instead of a class, a two-branch template for a three-outcome test, a 0.15 cutoff. The repair adopted here is structural: *every verdict is a printed comparison against the range of defensible references, never a boolean computed from a threshold.*

18 Falsification and contradiction handling

Stated before the claims are defended.

1. **F1 dies** if a response-blind elicitation shows $\cos(N_i, \text{own})$ far exceeding $\cos(N_i, \text{stranger})$. The present gap would then be an artefact of post-hoc writing rather than a property of elicitation.
2. **F2 dies** if a bounded M is exhibited that reproduces constraint behaviour across candidate sets. theorem 81 forbids it for unbounded S; a bounded-satisfaction variant with a stated bound would restrict the theorem's scope.
3. **F3 dies** if a fourth instrument reverses the operator ordering ($\rho < 0.8$). Three have not.
4. **F4 dies** if $\mathrm{Corr}(\text{influence}, \text{alignment}) \to 1$ under a different characteristic function. Only one has been tested; this is the weakest of the five.
5. **F5 dies** if the bit curve differs materially under another judge. Not tested.
6. **The framing itself dies** if a person's induced ordering is unstable across response sets. Then what was elicited is a property of the probe, not of the person, and nothing downstream measures values at all. This is the deepest available test and it requires two candidate sets per person, which this release does not have. corollary 71 (13.4% IIA violations) is a lower bound on the concern, not a resolution of it.

19 Discussion

19.1 The claim, and its novelty boundary

What is not new. Blackwell's theorem, Arrow's impossibility, Sen's liberal paradox, the Shapley value, Spearman–Brown, the design effect, attenuation from measurement error. The incompatibility between vetoes and scoring rules is classical social choice. Judge-sensitivity of language-model evaluation is documented. None of these is claimed here.

What is new, and only this. (i) The measured Δ distribution that turns the veto incompatibility into a *quantitative* statement about a specific release, with the counterexample present in the data (section 7.2). (ii) The self-versus-stranger decomposition showing that the individual signal in this elicitation is 0.083 of cosine (table 2). (iii) The comparative/absolute split under the gauge test (§9.3). (iv) The bit-rate of the weight scale (section 8.1). (v) The demonstration that the weighted-criteria score agrees with the Condorcet winner only 68.4% of the time while every ranking rule agrees 97–98%.

19.2 Why method relativity is not relativism

Every quantity here is relative to a declared query family (proposition 97), a declared decision family (definition 99), and a named instrument (§9). That is not an admission that the numbers are arbitrary; it is the statement that they are *typed*. A claim inherits the scope of the family it was computed against and may not be quoted outside it. The gauge test (§9.3) is what makes the typing empirical rather than rhetorical: it shows exactly which claims move when the instrument moves.

20 Limitations and epistemic status

"We did not run X" and "X cannot be run at this site" are different sentences and are not merged.

20.1 Structural — cannot be run here

1. **Response-blind source norms.** Every criterion was written after seeing the responses. proposition 111 shows contamination and redundancy are not separable without them.
2. **A second candidate set per person.** Required to test theorem 81 out of sample and to resolve the probe-dependence question of §18(6).
3. **Typed fields.** 82.3% of criteria carry none (table 3); scope and exception loss are unstudiable here.
4. **Compilation lineage.** Absent, so §8's three actions stay confounded.
5. Y**.** Not in the release.
6. **Cross-dataset / cross-domain.** One release exists.

20.2 Practical — could be run, was not

1. Held-out partitions for F1, F4, F5.
2. A specification curve over characteristic functions for the Shapley estimator; one was tested.
3. The bit curve of section 8.1 across judges.
4. Two-way clustering (theorem 55) on the person-level quantities; only prompt-level clustering was applied, and annotators repeat across prompts.

20.3 Quantitative bounds that apply to every number here

bound	value
judge attenuation ceiling	0.741
panel reliability ceiling	0.9722
compounded ceiling on any correlation	0.720
contamination-proxy variation in every flip rate	up to 1.8×
absolute winner claims: instrument share	≈ 40%
absolute winner claims: presentation-order share	8.0pp
rubric corpus never joined	1.8%
design shortfall for 1pp resolution	968 of 1,213 prompts
pairwise-dependence screen: MDE at full n	$\lvert\phi\rvert = 0.153$ against median 0.124

Remark 118 (The last row read honestly). The median pairwise association in the operator grid is *below* the MDE at the largest available sample. The pairwise table is therefore a screen for *strong* dependence and is worded that way throughout; its nulls are underpowered rather than informative.

21 What a design that can measure this requires

Each row is derived from a specific result above, not proposed.

requirement	derived from	unlocks
response-blind source norms	proposition 111	the only route separating contamination from redundancy
≥ 2 candidate sets per person	theorem 81, corollary 71	tests whether M transfers; quantifies candidate-set overfitting
typed fields: veto, scope, exception, priority as separate columns	§7, table 3	the index mismatch is a missing column, not a hard problem
interval-scale validation, or paired comparisons instead	table 1	11.7% of decisions currently rest on an unestablished assumption
$k > 4$ responses	theorem 64	raises the probe bound from 2 to $k - 2$
$\geq$ 1,213 prompts	corollary 37	1pp resolution
substantially more than 968 for pairwise structure	§20	the typical association is currently undetectable
$\geq$ 3 executors	§9.3	already available; the gauge is established
compilation receipt (lineage)	§8	separates selection, rewriting, weight-discarding
declared Q and declared A	proposition 97, remark 58	without them "loss" is undefined and "defect" is inseparable from "social choice"

22 Conclusion

The chain of equation (7) loses something at every arrow, and the losses are not of one kind.

- $\succeq_i \to N_i$**.** An unbounded relation is projected onto two free parameters (theorem 64) – forced, not recoverable. Within that, a person's own criteria recover 39% of the direction of their own choice, and the individual component of that is 0.083 of cosine. The type structure is empty: 82.3% of criteria carry no condition, exception, priority, hedge or prohibition. And 11.7% of decisions depend on an interval reading of a scale that was never validated as one.
- $N_i \to G$**.** Aggregation costs 0.1298 of normalised utility (theorem 74) – a social choice, but one made silently because A was never declared. The weighted-criteria route agrees with the Condorcet winner only 68.4% of the time while ranking rules agree 97–98%. 13.4% of pairwise relations flip when the option set changes. And 26.1% of the time the utilitarian winner is a response somebody called unacceptable.
- **Type.** No finite weight encodes a veto (theorem 81). This is the strongest statement in the paper and it required no instrument.
- $G \to C$**.** Compilation changes 29.4% of decisions, with its three actions confounded. But the discarded magnitudes were worth about one bit: sign alone reproduces 89.8%.
- $C \to D$**.** The executor's reliability is 0.5497, capping every correlation at 0.741. Changing the reader changes 40% of winners; reordering the page changes 8%. Comparative claims survive all of it; absolute claims do not.
- $D \to Y$**.** Does not exist in the release.

Theory claim. The losses inside the elicitation are larger than every loss the pipeline adds. The most defensible statement this object supports is a *theorem* about what a scalar scale cannot encode, and the most consequential one is that the next dataset should buy **fields and response-blindness**, not finer weights or more prompts.

References